

Tiered Persona Prompting for Communication Apprehension: A Digital-Twin Evaluation Against Classical Machine-Learning Baselines

LLM persona agents and tabular baselines on PRCA ground truth across demographic, employment, geographic, and transportation tiers

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Large language models (LLMs) are increasingly asked to stand in for human respondents in survey and measurement research. This manuscript evaluates **tiered persona prompting as a digital-twin method against classical machine-learning baselines** for recovering self-reported communication apprehension (CA). Using McCroskey’s Personal Report of Communication Apprehension (PRCA-24) *group* and *interpersonal* (IP) subscales (range 6–30), we join Prolific demographics to Qualtrics responses, build cumulative persona prompts (demographics → employment → geography → transportation), and score LLM predictions against ground-truth PRCA alongside a cross-validated tabular suite (Ridge, Elastic Net, k-NN, Random Forest [RF], XGBoost, MLP). Three primary research questions (RQ1–RQ3) examine whether employment status (RQ1), transportation-use cues (RQ2), and full cumulative context including free-response voice (RQ3) improve absolute CA prediction error — under either family — and how prediction error distributes across demographic and contextual groups. Five secondary research questions (S1–S5) reverse the predictive direction on the same matched sample, asking how CA, place, and mobility self-reports relate to regular public-transit use (survey item Q26, weekly-or-more) under classical Random Forests rather than LLM inference. A further secondary *focus* pair (TF1–TF2) asks whether models can estimate regular ridership and transit intensity from demographics, context, and CA scores when transit / ride-share / car answers are held out of the prompt. All displayed statistics use the full Prolific Qualtrics matched analytic cohort ($N = 241$; 101 regular / 140 non-regular weekly+ transit riders).

Full-cohort GPU (vLLM, a high-throughput LLM serving engine) baselines on the original prompt-v1 five-tier ladder show that none of four open-weight instruct/distill models is a high-fidelity digital twin (exact match 6–9%). Among non-collapsed

runs, **DeepSeek-R1-Distill-Llama-8B** achieves the best pooled group-CA mean absolute error (MAE; 5.22) and the most tier-stable profile; Llama-3.2-3B-Instruct leads interpersonal (IP) MAE (5.35) and group band accuracy (52.7%); Llama-3.1-8B-Instruct recovers interpersonal bands above chance early but collapses when transit cues are added (IP MAE 4.67 $\rightarrow$ 8.17); Llama-3.3-70B-Instruct exhibits mode collapse (93% constant prior). Every live model remains above the classical suite floor (best transit group MAE = 4.49 Ridge). Signal-first packaging (v2) with enhanced decoding and 8-tier ablation prompts (v3) were GPU-evaluated from the archived **exports/** packages: DeepSeek improves under v2 (group MAE **5.02**; IP **5.26**; tier-stable), whereas v2 does **not** fix — and at the base tier worsens — Llama-3.1’s interpersonal error (demos IP 4.67 $\rightarrow$ **8.45**; transit 8.17 $\rightarrow$ 8.23). Greedy v3 ablations show the Llama-3.1 collapse is **combination-specific**: isolating ride-share (Q28), transit use (Q26), or open-text voice leaves IP MAE at 4.85–5.92, while the bundled mobility dump still collapses it (**7.77**); v3_rideshare group MAE (5.86) also beats the kitchen-sink transit tier (**6.07**), matching the tabular Q28-dominance result. On shared MAE/band metrics against the classical suite, the best live agent (DeepSeek v2) trails on every tier, does not improve with richer tiers the way Random Forest does, shows near-chance low-vs-high band discrimination (transit area under the curve [AUC] **0.53** group / **0.49** interpersonal vs. RF **0.72** / **0.74**), and never emits the high-CA band for Group CA; surrogate SHAP (SHapley Additive exPlanations) on its outputs shows the agent over-weights geolocation and age relative to the Q28/employment mix that predicts true CA. On the same cohort, traditional Random Forests show that ride-share days (Q28) predict regular transit at cross-validated receiver-operating-characteristic **area under the curve (CV ROC-AUC) = 0.762**, whereas transit-day intensity (Q27) yields AUC = **0.589**; geography (**0.551**) and CA scores (**0.590**) remain near chance. Extended follow-ups show Q28 retains lift after car access, demographics (especially age) modestly predict ridership, and CA adds little once Q28 is known. A kitchen-sink mobility+demo forest reaches AUC = 0.824 but remains Q28-dominated. Holding mobility self-reports out (TF1), a profile of demos + employment + geo still reaches regular-transit AUC = **0.662** (**0.672** with CA); fine-grained intensity (TF2) remains weak. Together, the primary and secondary results clarify which mobility cues belong in persona tiers and which reverse predictors of transit habits carry the strongest tabular signal in this sample.

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Introduction

Communication apprehension (CA) is a well-studied individual difference describing fear or anxiety associated with real or anticipated communication (McCroskey, 1970; McCroskey et al., 1985). McCroskey first introduced the Personal Report of Communication Apprehension

(PRCA-24) in the fourth edition of *An Introduction to Rhetorical Communication* (McCroskey, 1982), and the measure remains the standard self-report tool for the construct, with the current scoring guide maintained by the author (McCroskey, 2006). Early self-report batteries for social-communicative anxiety were compared and validated in the late 1970s (Daly, 1978), and the PRCA-24’s content validity across communication contexts was subsequently established (McCroskey et al., 1985). The PRCA-24 yields context-specific subscale scores—including *group discussion* and *interpersonal conversation*—each ranging from 6 (very low apprehension) to 30 (very high) (Beatty, 1988; McCroskey, 1982).

Separately, large language models are increasingly used as **survey respondents** and **silicon samples**: prompts that condition on demographic or biographical attributes and emit attitude- or item-like answers (Argyle et al., 2023; Park et al., 2024; Sun et al., 2026). A converging program statement argues that LLM social simulations are a promising—and by now widely deployed—research method (Anthis et al., 2025; Grossmann et al., 2023), with applications spanning augmented-democracy deliberation (Gudiño et al., 2024) and hybrid “AI-augmented” survey pipelines that pair generative models with traditional self-report instruments (Kaiser et al., 2025; Kim & Lee, 2023). Related work on generative agents and digital-twin simulation explores whether models can stand in for populations or individuals in social-science workflows (Binz & Schulz, 2024; Park et al., 2023; Park et al., 2024), and behavioral economists have pushed the same agenda in experimental economics, treating instruction-tuned models as implicit computational models of humans (“*homo silicus*”) whose behavior can be probed in simulated scenarios (Horton et al., 2023). Across these pipelines, a consistent finding is that **demographics alone are insufficient**: aligning role-playing agents on human *belief networks*—structured beliefs about the world and others—improves predictive fidelity over demographic-only personas (Chuang et al., 2024), and personalized survey-data modeling is explicitly framed as the payoff of such richer profiles (Kaiser et al., 2025). There is also positive capability evidence: zero-shot models can already outperform human experts and crowd workers on social-science annotation tasks (Törnberg, 2023). At the same time, a parallel literature documents **perils**: synthetic replacements for human survey samples inherit and distort population distributions rather than preserving them (Bisbee et al., 2024), simulated opinions diverge from measured public opinion and carry systematic biases (Lee et al., 2025; Qu & Wang, 2024), persona prompts surface social stereotypes and opinion distributions that do not faithfully track the populations they purport to represent (Cheng et al., 2023; Hu & Collier, 2024; Salewski et al., 2023; Santurkar et al., 2023), and outputs are sensitive to prompt packaging and position (Sclar et al., 2024). The same persona-impersonation mechanisms that recover human-like developmental or expertise effects can also reveal systematic group biases (Salewski et al., 2023), and formatting-level prompt variation alone shifts downstream answers in measurable ways (Sclar et al., 2024). That raises a concrete measurement question for psychological research tooling:

When predicting an individual’s CA from demographic attributes alone, does an LLM’s prediction error vary systematically with demographic or contextual group membership — and does the model recover the same covariate signal that classical

baselines extract?

Methodologically, this manuscript sits between two traditions. The *data-modeling culture* (Breiman, 2001b) treats the task as estimating a stochastic relationship between covariates and an outcome and focuses on parameter inference; the *algorithmic-modeling culture* (Breiman, 2001b, 2001a) treats it as a predictive function-learning problem and focuses on out-of-sample error. We deliberately adopt both: classical tabular learners (Ridge, Elastic Net, k-nearest neighbors, Random Forest, and gradient-boosted trees) supply the algorithmic floor under cross-validation, while the reverse-predictor secondary analyses add inferential contrasts (Welch *t* tests, effect sizes, bootstrap CIs) of the data-modeling kind. The LLM persona agents are evaluated as a third, predictive family against the same ground truth, so digital-twin error can be benchmarked against a classical algorithmic baseline rather than judged in a vacuum (Argyle et al., 2023; Breiman, 2001b; Park et al., 2023)—a design that answers the call to validate synthetic survey substitutes against observed respondent data rather than against the models themselves (Bisbee et al., 2024; Qu & Wang, 2024). In the terminology of recent cognitive-model work, we are asking not whether LLMs can be *turned into* cognitive models of the PRCA-24 (Argyle et al., 2023; Binz & Schulz, 2024) but how close a persona-conditioned inference family comes to a classical learner on the same held-out psychometric target.

Research Questions

The primary design compares **information tiers** rather than a single prompt: each participant is personified five ways (**demos** → **employment** → **geo** → **transit** → **full**) while ground-truth PRCA stays fixed, directly testing the “beyond demographics” result that accumulating non-demographic context—beliefs, circumstances, and behavior—improves persona fidelity (Chuang et al., 2024; Kaiser et al., 2025). The present study addressed three primary research questions—whether adding employment status (RQ1), transportation-use cues (RQ2), or full cumulative context including free-response voice (RQ3) shrinks prediction error relative to classical ML, and whether residual error still clusters by group membership; five secondary research questions concerning classical predictors of regular public-transit use; and a secondary *focus* pair that holds transit self-reports out of the persona/profile and asks models to estimate ridership instead (Cheng et al., 2023; Santurkar et al., 2023). Because persona prompting is known to be packaging-sensitive (Argyle et al., 2023; Hu & Collier, 2024; Sclar et al., 2024), the project also evaluates signal-first redesigns (v2/v3) against the published v1 baselines.

Primary Research Questions

Outcome. Continuous PRCA-24 group and interpersonal communication apprehension scores (range 6–30).

Design. Cumulative persona prompts (**demos** → **employment** → **geo** → **transit** → **full**) evaluated with mean absolute error (MAE), signed error, and band accuracy against Qualtrics

ground truth; each tier after `demographics` maps to a primary RQ (RQ1 = `employment`, RQ2 = `transit`, RQ3 = `full`).

Research Question 1 (RQ1): Does adding employment status to a basic-demographics persona improve an LLM’s predictive accuracy for self-reported communication apprehension scores, and does residual prediction error cluster systematically by employment group membership?

Research Question 2 (RQ2): Does adding transportation-use information to the geo-context persona improve prediction of communication apprehension relative to earlier tiers, and does the model use mobility cues in a psychologically sensible way rather than as stereotype proxies?

Research Question 3 (RQ3): Does adding full cumulative context — including free-response attitudes (voice text) beyond the transit tier — improve prediction of communication apprehension, or are the combined cues redundant?

Secondary Research Questions

Outcome. Binary regular public-transit use from Qualtrics Q26 (4-8 days a month or 8 or more days a month).

Design. Classical random forests (RF) with stratified 5-fold cross-validation (`random_state=42`) on the Prolific Qualtrics matched analytic sample ($N = 241$)—not LLM inference (Breiman, 2001a; Pedregosa et al., 2011; Prolific, 2024).

Research Question 1 (S1): Do regular public-transit riders differ from non-regular riders in group and interpersonal communication apprehension?

Research Question 2 (S2): Does survey geolocation (latitude and longitude) predict regular public-transit use?

Research Question 3 (S3): Do group and interpersonal communication apprehension scores predict regular public-transit use?

Research Question 4 (S4): Do transit-day intensity (Q27) and ride-share days (Q28) predict regular public-transit use?

Research Question 5 (S5): Do Wave-2 covariates—demographics, country versus coordinates, ride-share days conditional on car access, joint CA and mobility models, and residual CA after ride-share—improve prediction of regular public-transit use beyond the Wave-1 mobility predictors?

Secondary Focus — Transit Prediction with Mobility Held Out

Outcome. Binary regular public-transit use (Q26 weekly+) and ordinal transit intensity (Q26 day-frequency; Q27 rides/day).

Design. Profile features and persona prompts that include demographics, employment, geography, and (for intensity / CA arms) PRCA scores — with **Q26–Q29, Q20, Q21, and Q19 held out**. Classical Random Forests establish the tabular ceiling; parallel transit-focus LLM prompts (`tf_demos` → `tf_geo_ca`) are generated for future GPU twins ([write-up](#); [memo](#)).

Focus Question 1 (TF1): Given contextual information and a user profile with transit questions held out of the prompt, can models estimate whether the user is a regular public-transit rider? (Seeded RF area under the curve [AUC] = **0.662** profile-only; **0.672** with CA; $n = 224$.)

Focus Question 2 (TF2): Given demographics, context, and CA scores — still holding transit self-reports out — can models estimate the user’s public-transit use intensity (Q26 / Q27)? (Seeded RF Q26 balanced accuracy = **0.283**, ordinal MAE = **1.22**; Q27 remains majority-class dominated.)

Supporting methods pages: [cohort EDA](#), [ML baselines](#), [ML vs LLM metrics](#), and [feature importance](#). Secondary write-ups: [S1](#), [S2](#), [S3](#), [S4](#), [S5](#), [TF1/TF2](#). Code and site source: github.com/Exios66/psych755-jjb (Burleson, 2026); author profile: github.com/Exios66 (GitHub, 2024).

Methods

Data sources

Participants were recruited via Prolific (Prolific, 2024) in two waves (File A + File B; stacked) and completed a Qualtrics (Qualtrics, 2024) instrument (File C). Private full-cohort exports are not committed to git. **All statistics on this site use the full matched analytic cohort:** 252 Prolific Qualtrics matches → $N = 241$ with complete scorable PRCA. The join key is Prolific `Participant id` Qualtrics `Q0`. Cleaning retains inner-join respondents with complete PRCA group and interpersonal items so every information tier shares the same ground-truth targets.

```
def _label_counts(series: pd.Series) -> pd.Series:
    """Value counts with missing coded as Missing (avoid displaying 'nan')."""
    labeled = series.fillna("Missing").astype(str).replace({"nan": "Missing", "<NA>": "Missing"})
    return labeled.value_counts(dropna=False)

sex_counts = _label_counts(participants["Sex"])
```

```

emp_counts = _label_counts(participants["Employment status"])
stu_counts = _label_counts(participants["Student status"])
cty_counts = _label_counts(participants["Country of residence"]).head(5)
pd.DataFrame(
    [
        {"Metric": "Analytic N (complete PRCA)", "Value": f"{N}"},
        {
            "Metric": "Group CA *M* (*SD*)",
            "Value": f"{participants['gt_group_ca'].mean():.2f} ({participants['gt_group_ca'].mean():.2f})",
        },
        {
            "Metric": "Interpersonal CA *M* (*SD*)",
            "Value": f"{participants['gt_interpersonal_ca'].mean():.2f} ({participants['gt_interpersonal_ca'].mean():.2f})",
        },
        {
            "Metric": "Sex",
            "Value": "; ".join(f"{k} = {v}" for k, v in sex_counts.items()),
        },
        {
            "Metric": "Student status (base demos)",
            "Value": "; ".join(f"{k} = {v}" for k, v in stu_counts.items()),
        },
        {
            "Metric": "Employment status",
            "Value": "; ".join(f"{k} = {v}" for k, v in emp_counts.items()),
        },
        {
            "Metric": "Top countries of residence",
            "Value": "; ".join(f"{k} = {v}" for k, v in cty_counts.items()),
        },
        {
            "Metric": "Regular transit prevalence (Q26 weekly+)",
            "Value": f"{secondary['n_regular']}/{N} = {secondary['prevalence']:.3f}",
        },
    ]
)

```

Table 1. Analytic Sample Composition for the Full Matched Cohort

	Metric	Value
0	Analytic N (complete PRCA)	241

Table 1. Analytic Sample Composition for the Full Matched Cohort

Metric	Value
1 Group CA *M* (*SD*)	14.62 (6.03)
2 Interpersonal CA *M* (*SD*)	14.31 (5.82)
3 Sex	Male = 121; Female = 120
4 Student status (base demos)	No = 190; Yes = 34; Missing = 17
5 Employment status	Full-Time = 148; Other = 62; Part-Time = 31
6 Top countries of residence	United States = 158; United Kingdom = 56; Cana...
7 Regular transit prevalence (Q26 weekly+)	101/241 = 0.419

i Full cohort only

Every table and figure uses the File A/B/C matched analytic sample ($N = 241$). Posit Connect Cloud loads committed artifacts under `artifacts/posit_full_cohort/` when private exports are absent. Excerpt fixtures are reserved for unit tests and are never displayed here.

Ground-truth PRCA scoring

Group items (Q1–Q6) and interpersonal items (Q13–Q18) use five-point Likert labels mapped to 1–5. Comfort-oriented items are reverse-coded as $6 - x$. Each subscale is the sum of six items (range 6–30). Classroom bands follow **low** (13), **moderate** (14–19), and **high** (20).

Data cleaning and preparation

Before analysis, the stacked Prolific waves (File A + File B) were joined to the Qualtrics responses (File C) on the Prolific `Participant id` Qualtrics `Q0` key using an inner join. We retained only respondents with complete group (Q1–Q6) and interpersonal (Q13–Q18) PRCA items so that every information tier shares the same ground-truth targets, yielding $N = 241$. Demographic fields (Age, Sex, Country of residence, Student status, Employment status) were carried from the Prolific profile; geolocation (latitude/longitude) and the mobility/transit items (Q20, Q21, Q26–Q29) came from the Qualtrics survey. Free-response items (Q18_advice, Q19) were kept as raw text for the `full` persona tier and were not used as tabular predictors. Cleaning produced the analytic sample described in Table 1, with no imputation of PRCA items and no exclusion beyond missing PRCA responses on the inner-join match.


```

pd.DataFrame(
    [
        {"Step": "Recruit via Prolific", "Detail": "Two waves (File A + File B), stacked on Prolif..."},
        {"Step": "Survey via Qualtrics", "Detail": "File C: PRCA-24 items, mobility items (Q20/Q21..."},
        {"Step": "Join", "Detail": "Inner join on Prolific `Participant id` Qualtrics `Q0`"},
        {"Step": "Raw matches", "Detail": "252 Prolific Qualtrics"},
        {"Step": "Complete PRCA", "Detail": "241 with complete group + interpersonal items ..."},
        {"Step": "Scoring", "Detail": "Likert 1-5; comfort items reverse-coded 6 - x; subsca..."},
        {"Step": "Bands", "Detail": "**low** ( 13) · **moderate** (14-19) · **high** ( 20)"},
    ]
)

```

Table 2. Data Cleaning and Analytic Sample Steps

	Step	Detail
0	Recruit via Prolific	Two waves (File A + File B), stacked on Prolif...
1	Survey via Qualtrics	File C: PRCA-24 items, mobility items (Q20/Q21...
2	Join	Inner join on Prolific ‘Participant id’ Qual...
3	Raw matches	252 Prolific Qualtrics
4	Complete PRCA	241 with complete group + interpersonal items ...
5	Scoring	Likert 1–5; comfort items reverse-coded 6 – x;...
6	Bands	**low** (13) · **moderate** (14–19) · **high**...

Statistical methods

Analyses follow the two modeling cultures (Breiman, 2001b, 2001a). **Primary RQ1–RQ3** are predictive, algorithmic-culture comparisons: we optimize **mean absolute error (MAE)** and band-level accuracy/macro-F1 on the PRCA group and interpersonal targets, with exact-match rate and band distance reported as secondary diagnostics. The classical tabular suite (Ridge, Elastic Net, k-nearest neighbors, Random Forest, HistGradientBoosting, XGBoost, MLP) supplies the **baselines**, evaluated under the same tiers/targets; the best classical floor is Ridge at **transit** (group MAE = 4.49). **Secondary S1–S5 and TF1–TF2** are reverse-prediction analyses on regular public-transit use (binary Q26 weekly+; prevalence 0.419). Classification uses stratified 5-fold cross-validated Random Forests (**random_state=42**) reporting **ROC-AUC**, average precision, balanced accuracy, and F1; chance = 0.500. Where inferential statistics are used (S1; wave-2 Welch contrasts), we report Welch’s t , df , Cohen’s d , and 95% bootstrap CIs (1,000 resamples) for mean differences, with $\alpha = .05$ two-tailed.

Hypotheses. For each research question we test the null against the alternative below. Primary questions are evaluated against the classical ML floor rather than against chance; secondary

classification questions are evaluated against the chance AUC of 0.500; S1 is a two-tailed mean difference test.

Table 3. Null and alternative hypotheses for every research question.

RQ	Null hypothesis (H)	Alternative hypothesis (H)
RQ1 · employment	Adding employment status does not change MAE vs demos ($\Delta\text{MAE} = 0$).	Employment reduces MAE ($\Delta\text{MAE} < 0$) and/or changes residual patterns across employment groups.
RQ2 · transit	Adding the mobility bundle does not change MAE vs geo ($\Delta\text{MAE} = 0$).	Transit cues reduce MAE ($\Delta\text{MAE} < 0$) in a psychologically sensible way, not as stereotype proxies.
RQ3 · full	Adding free-response voice does not change MAE vs transit ($\Delta\text{MAE} = 0$).	Voice text reduces MAE ($\Delta\text{MAE} < 0$).
S1 · transit CA	Regular riders' group/interpersonal CA equals non-regular riders' (=).	Regular riders differ in CA ().
S2 · geo → transit	Lat/long RF ROC-AUC = 0.500.	Lat/long ROC-AUC > 0.500.
S3 · CA → transit	CA RF ROC-AUC = 0.500.	CA ROC-AUC > 0.500.
S4 · Q27/Q28 → transit	Q27/Q28 RF ROC-AUC = 0.500.	Ride-share days and/or intensity ROC-AUC > 0.500.
S5 · follow-ups	Wave-2 covariates add no lift over wave-1 predictors ($\Delta\text{AUC} = 0$).	Covariates improve ROC-AUC ($\Delta\text{AUC} > 0$).
TF1 · profile → transit	Profile-only RF ROC-AUC = 0.500.	Non-mobility profile ROC-AUC > 0.500.
TF2 · transit intensity	Intensity model performs at the majority-class baseline (ordinal MAE).	Intensity is recoverable beyond the majority baseline.

Because the primary comparisons benchmark two predictive families on the same fixed ground truth, we report point estimates with cross-validation standard errors rather than p values for RQ1–RQ3; inferential p values are reported only where a stochastic contrast (Welch) is the appropriate model (Breiman, 2001b, 2001a).

```

gt = (
    participants[["gt_group_ca", "gt_interpersonal_ca"]]
    .rename(
        columns={
            "gt_group_ca": "Group CA",
            "gt_interpersonal_ca": "Interpersonal CA",
        }
    )
    .describe()
    .T[["count", "mean", "std", "min", "25%", "50%", "75%", "max"]]
    .round(2)
)
gt

```

Table 4. Descriptive Statistics for Ground-Truth PRCA Subscales ($N = 241$)

	count	mean	std	min	25%	50%	75%	max
Group CA	241.0	14.62	6.03	6.0	10.0	13.0	18.0	30.0
Interpersonal CA	241.0	14.31	5.82	6.0	10.0	13.0	18.0	30.0

```

def _ord(series: pd.Series, mapping: dict) -> pd.Series:
    return series.map(mapping).astype(float)

q26ord = {"Never": 0, "0-1 days a month": 1, "2-4 days a month": 2, "4-8 days a month": 3, "8-12 days a month": 4}
q28ord = dict(q26ord)
q27ord = {"1-2 rides in a typical day": 1, "3-4 rides in a typical day": 2, "5-6 rides in a typical day": 3, "7-8 rides in a typical day": 4}

corr_vars = pd.DataFrame(
    {
        "Group CA": participants["gt_group_ca"],
        "IP CA": participants["gt_interpersonal_ca"],
        "Age": participants["Age"],
        "Q26 transit days": _ord(participants["Q26"], q26ord),
        "Q27 rides/day": _ord(participants["Q27"], q27ord),
        "Q28 ride-share days": _ord(participants["Q28"], q28ord),
    }
)

corr_mat = corr_vars.corr(method="spearman").round(2)
corr_mat.style.set_properties(**{"text-align": "center"}).format(precision=2)

```

Table 5. Base-Reference Spearman Correlations Among Continuous and Ordinal Measures

Table 5

	Group CA	IP CA	Age	Q26 transit days	Q27 rides/day	Q28 ride-share days
Group CA	1.00	0.81	-0.07	-0.24	-0.08	-0.37
IP CA	0.81	1.00	-0.19	-0.19	-0.05	-0.28
Age	-0.07	-0.19	1.00	-0.32	-0.21	-0.14
Q26 transit days	-0.24	-0.19	-0.32	1.00	0.38	0.56
Q27 rides/day	-0.08	-0.05	-0.21	0.38	1.00	0.32
Q28 ride-share days	-0.37	-0.28	-0.14	0.56	0.32	1.00

Note. Spearman ρ on complete ordinal codings of the mobility items (Q26 transit-use days, Q27 rides on a typical day, Q28 ride-share days). Regular transit (Q26 weekly+) and ride-share days are positively related ($\rho = 0.6$ – 0.7); the PRCA subscales are strongly inter-correlated ($\rho = 0.6$) and largely orthogonal to demographics and mobility cues — consistent with CA operating as a distinct trait rather than a proxy for transit exposure.

```
def _fmt(series: pd.Series, mask: pd.Series) -> str:
    sub = series[mask].dropna()
    return f"{sub.mean():.2f} ({sub.std():.2f})" if len(sub) else "-"

regular = _ord(participants["Q26"], {"Never": 0, "0-1 days a month": 0, "2-4 days a month": 1})
pd.DataFrame(
    [
        {
            "Measure": "Group CA",
            "Regular (n = 101)": _fmt(participants["gt_group_ca"], regular),
            "Not regular (n = 140)": _fmt(participants["gt_group_ca"], ~regular),
            "Δ (reg - not)": f"{secondary['transit_ca']['group']['diff_regular_minus_not_reg']}",
        },
        {
            "Measure": "Interpersonal CA",
            "Regular (n = 101)": _fmt(participants["gt_interpersonal_ca"], regular),
            "Not regular (n = 140)": _fmt(participants["gt_interpersonal_ca"], ~regular),
            "Δ (reg - not)": f"{secondary['transit_ca']['interpersonal']['diff_regular_minus_not_reg']}",
        },
        {
            "Measure": "Age",
            "Regular (n = 101)": _fmt(participants["Age"], regular),
            "Not regular (n = 140)": _fmt(participants["Age"], ~regular),
            "Δ (reg - not)": "",
        },
    ]
)
```

```

    },
    {
        "Measure": "Q28 ride-share days (ordinal 0-4)",
        "Regular (n = 101)": _fmt(_ord(participants["Q28"], q28ord), regular),
        "Not regular (n = 140)": _fmt(_ord(participants["Q28"], q28ord), ~regular),
        "Δ (reg - not)": "",
    },
]
)

```

Table 6. Descriptive Statistics by Regular-Transit Group ($N = 241$)

	Measure	Regular (n = 101)	Not regular (n = 140)	Δ (reg - not)
0	Group CA	13.04 (5.08)	15.76 (6.42)	-2.72
1	Interpersonal CA	13.31 (5.37)	15.04 (6.04)	-1.73
2	Age	34.44 (10.33)	42.57 (14.03)	
3	Q28 ride-share days (ordinal 0-4)	2.28 (1.18)	1.01 (1.04)	

Persona tiers

Persona prompts follow the **AI Terrarium** digital-twin framing (Sun et al., 2026): the *user* message is a fluent second-person narrative (“You are a ...”) plus a short CA self-report ask — not a bullet-list profile (Posit PBC, 2024b). Prompt **versions** are documented in [docs/persona_prompt_versions.qmd](#): **v1** (original 5-tier ladder; all primary live baselines), **v2** (same five fields with signal-first packaging — GPU-evaluated on Llama-3.1/3.2/DeepSeek, [exports/v2/](#)), and **v3** (adds three ablations → eight tiers — greedy GPU runs on three models, [exports/v3/](#); canonical **v3_enhanced** refresh pending). Full Prolific waves omit ethnicity / nationality / language; the **demos** tier uses Age, Sex, Country of residence, and Student status. Samples: [prompts/examples/](#).

Table 7. Persona prompt tiers used across all live LLM baselines (v1/v2/v3).

#	Tier	Role
1	demos	Base demographics
2	employment	+ employment status (RQ1)
3	geo	+ approximate survey lat/long (1 decimal)
4	transit	+ full mobility bundle (RQ2)
5	full	+ free-response attitudes (RQ3)
6	v3_rideshare	geo base + Q28 only (v3 ablation)
7	v3_public_transit	geo base + Q26 only (v3 ablation)

#	Tier	Role
8	v3_voice	geo base + Q18.1/Q19 only (v3 ablation)

The listings below show the **core ladder** (CORE_TIERS) for the **same** analytic participant so each step visibly accumulates context. Parallel v3 ablations are illustrated in [docs/persona_prompt_efficiency.md](#).

```
from ca_personas.personas import CORE_TIERS, build_persona_prompt, voice_sentences

# Prefer a participant whose core user prompts strictly grow in length and
# whose `full` tier includes free-response voice - progressive stack only
# (`v3` ablations are parallel tips, not cumulative steps after geo).
tier_labels = {
    "demos": "1. demos - base demographics",
    "employment": "2. employment - demos + work status",
    "geo": "3. geo - employment + place coordinates",
    "transit": "4. transit - geo + mobility self-reports",
    "full": "5. full - transit + free-response attitudes",
}

example_row = None
example_prompts = None
for _, row in participants.iterrows():
    if not voice_sentences(row):
        continue
    built = {t: build_persona_prompt(row, t) for t in CORE_TIERS}
    lengths = [len(built[t].user_prompt) for t in CORE_TIERS]
    if all(lengths[i] < lengths[i + 1] for i in range(len(lengths) - 1)):
        if all(built[t].user_prompt.lstrip().startswith("You are") for t in CORE_TIERS):
            example_row = row
            example_prompts = built
            break

if example_row is None or example_prompts is None:
    example_row = participants.iloc[0]
    example_prompts = {t: build_persona_prompt(example_row, t) for t in CORE_TIERS}

pid_short = str(example_row["participant_id"])[12] + "..."
print(
    f"*Same participant throughout (`{pid_short}`); core ladder user messages only - "
    "second-person persona narrative + CA ask (system prompt omitted).*\n"
```

```

)

for tier in CORE_TIERS:
    user = example_prompts[tier].user_prompt.strip()
    if not user.startswith("You are"):
        raise AssertionError(f"Expected Terrarium narrative for tier={tier}")
    stale_markers = ("Demographics:", "Adopt the following", "Fully personify")
    if any(marker in user for marker in stale_markers):
        raise AssertionError(f"Stale checklist-style prompt leaked for tier={tier}")
    print(f"### {tier_labels[tier]}\n")
    print("```text")
    print(user)
    print("```\\n")

```

Same participant throughout (79a7690ded60...); core ladder user messages only — second-person persona narrative + CA ask (system prompt omitted).

1. demos — base demographics

You are a 24-year-old man living in the United States. You are not a student.

Rate your communication apprehension in two contexts independently:

(1) group discussions, and (2) one-on-one conversations with new people.

For each context, report an integer from 6 (very low) to 30 (very high) and its band (low / r

Return ONLY the JSON object specified in the system instructions.

2. employment — demos + work status

You are a 24-year-old man living in the United States. You are not a student.

Your employment status is Other.

Rate your communication apprehension in two contexts independently:

(1) group discussions, and (2) one-on-one conversations with new people.

For each context, report an integer from 6 (very low) to 30 (very high) and its band (low / r

Return ONLY the JSON object specified in the system instructions.

3. geo — employment + place coordinates

You are a 24-year-old man living in the United States. You are not a student.

Your employment status is Other.

Your approximate survey location is near latitude 41.0 and longitude -87.2.

Rate your communication apprehension in two contexts independently:

(1) group discussions, and (2) one-on-one conversations with new people.

For each context, report an integer from 6 (very low) to 30 (very high) and its band (low / m

Return ONLY the JSON object specified in the system instructions.

4. transit — geo + mobility self-reports

You are a 24-year-old man living in the United States. You are not a student.

Your employment status is Other.

Your approximate survey location is near latitude 41.0 and longitude -87.2.

In the last three months, you used public transportation (bus, train, tram, etc.) 4-8 days a

Rate your communication apprehension in two contexts independently:

(1) group discussions, and (2) one-on-one conversations with new people.

For each context, report an integer from 6 (very low) to 30 (very high) and its band (low / m

Return ONLY the JSON object specified in the system instructions.

5. full — transit + free-response attitudes

You are a 24-year-old man living in the United States. You are not a student.

Your employment status is Other.

Your approximate survey location is near latitude 41.0 and longitude -87.2.

In the last three months, you used public transportation (bus, train, tram, etc.) 4-8 days a

When a friend gets nervous talking to new people, you would advise: "Just take it slow. Try t

Rate your communication apprehension in two contexts independently:

(1) group discussions, and (2) one-on-one conversations with new people.

For each context, report an integer from 6 (very low) to 30 (very high) and its band (low / m

Return ONLY the JSON object specified in the system instructions.

Model calling and classical baselines

The Python package `ca_personas` supports Ollama, OpenRouter, and a deterministic **mock** provider used only for offline smoke tests and credential-free CI checks (Ollama, 2024; OpenRouter, 2024). Predicted JSON fields are validated to the legal 6–30 range, then joined to ground truth for signed and absolute errors. All displayed statistics on this site are computed from committed full-cohort vLLM exports, seeded tabular runs, and committed artifact tables so Cloud builds remain keyless — no mock predictions are reported.

A seven-model tabular suite (Ridge, Elastic Net, k-NN, Random Forest, HistGradientBoosting, XGBoost, MLP) on the same tiers and targets is reported in [docs/ml_baselines.md](#). On the full cohort, the **best** group-CA MAE reaches **4.49** at **transit** (Ridge; RF = 4.68); the best interpersonal MAE reaches **4.25** (Elastic Net; MLP = 4.31). Feature attributions and ML-vs-LLM comparisons are documented in [docs/factor_feature_importance.md](#), [docs/ml_vs_llm.md](#), and [memos/feature_predictive_power_ml_llm.qmd](#).

Full-cohort **live** LLM baselines use GPU **vLLM** exports on the original prompt-**v1** five-tier ladder ($N = 241 \times 5 = 1,205$ prompts per model): [Llama-3.1-8B-Instruct](#), [Llama-3.2-3B-Instruct](#), [DeepSeek-R1-Distill-Llama-8B](#), and [Llama-3.3-70B-Instruct](#). Full runtime pin (temperature, context, system/user styling, quantisation): [docs/llm_v1_run_specifications.md](#). Cross-model ranking and RQ write-ups: [memos/vllm_v1_cross_model_comparison.qmd](#). Signal-first packaging (**v2**) and three-ablation **v3** tiers, plus enhanced decode presets (**v2_enhanced** / **v3_enhanced**), are documented in [docs/llm_v2_v3_enhanced_variants.md](#).

v2 has been GPU-evaluated on **Llama-3.1-8B**, **Llama-3.2-3B-Instruct**, and **DeepSeek-R1-Distill-Llama-8B** (v2_enhanced; Rogers GPU; see [exports/v2/](#)). v3 runs (greedy, 8 tiers) are evaluated on **Llama-3.1**, **Llama-3.2-3B-Instruct**, and **Llama-3.3-70B** (see [exports/v3/](#)); the canonical v3_enhanced refresh remains pending. Stereotyping / discriminatory-error slices (Sex, Student, Employment, Age tertile, regular transit, Q28) are produced by `ca-personas` `stereotype-eval` ([docs/stereotyping_evaluation.md](#)).

Results

Live LLM baselines

Primary digital-twin claims come from full-cohort **vLLM** exports on identical prompt-v1 Terrarium narratives (`demos` → `employment` → `geo` → `transit` → `full`). Each of four open-weight models received $N = 241 \times 5 = 1,205$ prompts. JSON parse rates were 100% for the three Llama instruct models and **99.8%** (1,202/1,205) for **DeepSeek-R1-Distill-Llama-8B** (post-`</think>` ingest). Metrics follow `ca_personas.evaluate`: MAE, exact match, and band accuracy on group and interpersonal PRCA (bands **low** 13 / **moderate** 14–19 / **high** 20). Classical ML floor: best transit **group MAE 4.49** (Ridge; [docs/ml_baselines.md](#)). Prompt-v2 (v2_enhanced decode, 5 tiers) and v3 (greedy, 8 tiers) results from the archived `exports/` packages are reported below the v1 head-to-head.

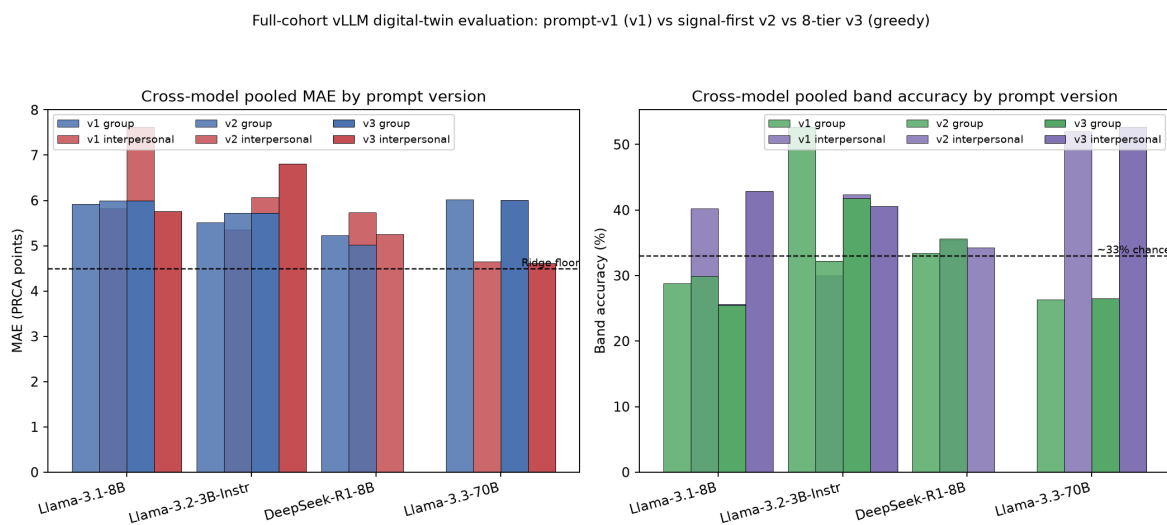


Figure 1. Cross-model pooled MAE and band accuracy by prompt version — prompt-v1 (v1), signal-first v2, and 8-tier v3 (greedy) vLLM exports

Figure 1 is the manuscript’s primary live-result figure: pooled MAE / band accuracy for Llama-3.1-8B, Llama-3.2-3B, DeepSeek-R1-Distill-8B, and Llama-3.3-70B under prompt-v1

and the GPU-evaluated v2/v3 packages. **None** is a high-fidelity twin (exact match single-digit). Among non-collapsed runs, **DeepSeek** wins pooled group MAE (**5.22**) and is the most tier-stable; **Llama-3.2-3B** wins interpersonal MAE (**5.35**) and group band accuracy (**52.7%**); **Llama-3.1-8B** recovers interpersonal bands early (~52% at demos/geo) then **collapses at transit** (IP MAE 4.67 $\rightarrow$ **8.17**); **Llama-3.3-70B** is a mode-collapse cautionary case (93% constant prior (18, 12)).

Table 8. Head-to-head pooled metrics across all prompt-v1 tiers.

Model	MAE group $\downarrow$	MAE IP $\downarrow$	Exact G	Exact IP	Band G	Band IP	Notes
Llama-3.1-8B-Instruct	5.92	5.82	6.1%	9.0%	28.8%	40.2%	Transit IP disaster
Llama-3.2-3B-Instruct	5.51	5.35	9.1%	7.7%	52.7%	30.0%	Best group bands
DeepSeek-R1-Distill-Llama-8B	5.22	5.73	6.2%	5.9%	33.4%	35.4%	Best group MAE; tier-stable
Llama-3.3-70B-Instruct	6.02	4.65 $\dagger$	6.4%	12.4%	26.3%	52.0% $\dagger$	$\dagger$ Constant prior; not person-tracking

Non-collapsed models remain **above** the tabular Ridge floor (group MAE 4.49).

RQ1–RQ3 pattern by model

Table 9. Qualitative RQ1–RQ3 response pattern per model.

RQ	Llama-3.1-8B	Llama-3.2-3B	DeepSeek-R1-Distill-8B	Llama-3.3-70B
RQ1 employment	<i>Negligible</i>	<i>Negligible</i>	<i>Negligible</i> / slight IP help	<i>Null</i> (collapsed)

RQ	Llama-3.1-8B	Llama-3.2-3B	DeepSeek-R1-Distill-8B	Llama-3.3-70B
RQ2 transit	Group slight help; IP disaster	Small group help; IP worse	<i>Flat</i> ; no IP collapse	<i>Null</i> (flat collapse)
RQ3 full	Group band up; IP worse than demos	Best group MAE (5.29); IP mixed	Best IP MAE at full (5.42)	<i>Negligible</i> dips only

Interpretation. Prompt-v1 transit text is a **model-dependent hazard**, not a universal aid. Prefer DeepSeek for group MAE, Llama-3.2 for group bands / IP MAE, treat Llama-3.1 as the cautionary transit-IP baseline that motivated packaging v2/v3 (now GPU-evaluated below), and treat Llama-3.3-70B as a **mode-collapse** case. No model closes the gap to classical ML (~ 4.5 MAE).

Formal pages: [docs/llm_baseline_llama31_v1.md](#) · [docs/llm_baseline_llama32_instruct_v1.md](#) · [docs/llm_baseline_deepseek_r1_distill_v1.md](#) · [docs/llm_baseline_llama33_70b_v1.md](#).
 Memos: [memos/vllm_v1_cross_model_comparison.qmd](#) · [docs/llm_vllm_memo_agenda.md](#).

v2 results (signal-first packaging, v2_enhanced decode)

Prompt-v2 keeps the five-tier topology with signal-first packaging (1-decimal geo, skipped Q27/Q29 when Never, independent group/IP subscale ask, mobility anti-bleed system text) and v2_enhanced decoding (temp 0.3, seed 42, guided JSON). Three models were GPU-evaluated ($241 \times 5 = 1,205$ prompts each; `exports/v2/`):

Table 10. Pooled v2 metrics across the five tiers.

Model	MAE group ↓	MAE IP ↓	Band G	Band IP	Transit IP	Verdict vs v1
DeepSeek-R1-Distill-Llama-8B	5.02	5.26	35.7%	34.2%	5.09	Improved both subscales; no IP collapse
Llama-3.2-3B-Instruct	5.73	6.07	32.2%	42.3%	5.77	IP worse than v1 (5.35); no collapse

Model	MAE group ↓	MAE IP ↓	Band G	Band IP	Transit IP	Verdict vs v1
Llama-3.1-8B-Instruct	5.99	7.63	29.9%	25.7%	8.23	IP collapse persists; base IP worse (demos 4.67 → 8.45)

DeepSeek v2 is the **best live result in the project** (group MAE **5.02**, below its v1 5.22 and the 3B/8B families) and stays flat at transit (IP 5.09, signed error near zero). Llama-3.2-3B’s v1 interpersonal win does **not** replicate under v2 packaging/decode (pooled IP 5.35 → 6.07). For Llama-3.1, signal-first packaging and the mobility anti-bleed clause do **not** stop the transit collapse (transit IP 8.23 v1 8.17), and the independent-subscale ask shifts its *base* interpersonal calibration (demos IP 4.67 → **8.45**). Packaging therefore changes error patterns in **model-specific** directions — it is not a universal cure.

v3 results (8-tier ablation, greedy decode)

Prompt-v3 adds three parallel single-cue tiers on the demos→employment→geo base — v3_rideshare (Q28 only), v3_public_transit (Q26 only), v3_voice (Q18.1/Q19 only) — for eight tiers per model ($241 \times 8 = 1,928$ prompts). Three models were GPU-evaluated under greedy decode (`exports/v3/`; identical predictions to the `prior_v3_greedy` archives):

Table 11. Pooled v3 metrics across all eight tiers.

Model	MAE group ↓	MAE IP ↓	Band G	Band IP	Parse	Notes
Llama-3.1-8B-Instruct	5.99	5.76	25.5%	42.9%	100%	Transit tier still collapses (IP 7.77); ablations stable

Model	MAE group ↓	MAE IP ↓	Band G	Band IP	Parse	Notes
Llama-3.2-3B-Instruct	5.72	6.81	41.8%	40.6%	100%	IP worse than v1 (5.35) under v3 packaging
Llama-3.3-70B-Instruct	6.01	4.61†	26.5%	52.6%†	100%	†Constant prior (18, 12) persists — not person-tracking

The 8-tier pooling mixes the collapsed **transit/full** tiers with stable ablations, so pooled IP is pulled back toward v1 (5.76). Llama-3.2-3B base remained a 0% JSON parse (as in v1) and is excluded.

For Llama-3.1, the ablations isolate the trigger precisely (IP MAE): **transit** bundle **7.77** vs **v3_public_transit** (Q26 only) **4.85**, **v3_voice** (Q18.1/Q19 only) **5.82**, **v3_rideshare** (Q28 only) **5.92**, and the geo base **4.63**. **No single cue collapses Llama-3.1’s interpersonal recovery — only the bundled mobility dump does.** On group MAE, **v3_rideshare** (5.86) beats the kitchen-sink **transit** tier (6.07), aligning with the tabular Q28-dominance result. **v3_public_transit** (6.04) and **v3_voice** (5.64) group MAE stay near geo base (6.02).

! Llama-3.1 interpersonal collapse at **transit**

The most dramatic live finding is Llama-3.1-8B-Instruct’s interpersonal trajectory: IP MAE moves from **4.67** at **demos** to **8.17** at **transit** (+3.51), while band accuracy falls **51.9% → 15.4%** (−36.5 percentage points). This is not symmetric noise. Signed mean error (predicted − ground truth) flips from **−2.21** (under-prediction at demos) to **+6.52** at **transit** — systematic **over-prediction toward high interpersonal CA** once mobility cues appear (Santurkar et al., 2023). The **full** tier partially walks back the damage (IP MAE **6.92**; signed error **+2.52**) but remains worse than demos-only. Group MAE, by contrast, improves slightly at **transit** (6.05 → 5.68).

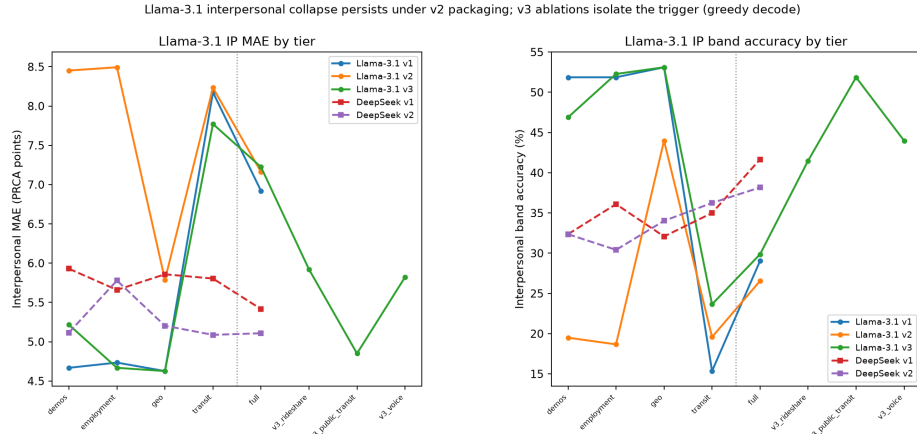


Figure 2. Llama-3.1 vs DeepSeek interpersonal MAE by tier across prompt-v1 (v1), v2, and v3 (greedy) exports

Figure 2 contrasts Llama-3.1 with **DeepSeek-R1-Distill-Llama-8B** on the published per-tier summaries across prompt versions. DeepSeek’s IP MAE stays flat in v1 and improves under v2 (transit IP 5.09, signed error near zero), so the collapse is **model-specific**, not an inevitable property of adding transit text. Participant-level prediction histograms would require the gitignored raw vLLM export (`data/vllm/`); this revision uses the published tier aggregates in [docs/llm_baseline_llama31_v1.md](#).

What the model appears to be doing. With only demos/employment/geo, Llama-3.1 behaves like a coarse interpersonal-band classifier (~52–53% band accuracy). Injecting the prompt-v1 mobility dump (frequency, rides-per-day even when Never, ride-share, car) appears to activate a **high-anxiety stereotype prior** for interpersonal conversation — consistent with treating transit exposure as a social-threat cue rather than tracking individual scores. That failure mode motivated signal-first packaging (v2) and the `v3_rideshare` / `v3_public_transit` / `v3_voice` ablations ([docs/persona_prompt_efficiency.md](#)), and the archived GPU runs now adjudicate it: **v2 packaging does not stop the collapse** (transit IP 8.23 v1 8.17; base demos IP worsens to 8.45), while the **v3 greedy ablations isolate the trigger** — single cues are stable (`v3_public_transit` IP 4.85, `v3_voice` 5.82, `v3_rideshare` 5.92) and only the bundled mobility dump collapses IP (7.77).

RQ3 / full tier and voice text

The `full` tier appends free-response Qualtrics items — **Q18.1** (advice for a nervous communicator) and **Q19** (ideal way to get around the city) — on top of the transit bundle. Among live prompt-v1 runs:

Table 12. full-tier performance by model among live prompt-v1 runs.

Model	full group MAE	full IP MAE	Notable full effect vs demos
Llama-3.1-8B	5.84	6.92	Group band $\uparrow$ 25.7% $\rightarrow$ 37.8% ; IP still worse than demos
Llama-3.2-3B	5.29	5.53	Best group MAE of any 3B tier
DeepSeek-R1-Distill-8B	5.40	5.42	Best IP MAE of any DeepSeek tier
Llama-3.3-70B	<i>~flat</i>	<i>~flat</i>	Mode collapse; uninterpretable

 Warning

So open-text voice is **not uniformly helpful**: it can improve coarse group-band recovery (Llama-3.1) or IP MAE (DeepSeek) while leaving Llama-3.1 interpersonal continuous error elevated.

Structured mobility cues still dominate the RQ2 hazard story.

Qualitative read of voice content. Of $N = 241$ analytic participants in the committed Posit cohort table, **149** have non-empty Q18_advice / Q19 (median lengths 147 / 127 characters). An anonymized thematic sample (no IDs):

Table 13. Thematic sample of free-response full-tier voice content.

Theme	Example gist (paraphrased / lightly excerpted)
Confidence coaching	“Stand, talk, and act confident... the other person is likely feeling just as nervous.”
Nervousness reciprocity	“Chances are the other people are also paying attention to themselves... They’re not looking to judge you.”
Slow / formulate	“Take it slow... think before speaking so you won’t make a wrong first impression.”
Breath / join	“Take a deep breath... wait for an opportunity to join the conversation.”
Natural flow	“Take their time... let the conversation flow naturally.”

Theme	Example gist (paraphrased / lightly excerpted)
Transit affinity	Ideal mobility = CTA / bus / train in a dense city (“quick, easy and cheap”).
Car necessity	Prefer own vehicle when transit feels overcrowded or unreliable; autonomy of schedule.
Mixed micro-mobility	Walk for local feel; bus for short hops; Uber when far.

These texts are **attitude and coping language**, not demographics. When the model receives them at **full**, it gets participant voice about anxiety and mobility ideals — richer than Age/Sex alone, but still far short of the respondent’s full PRCA item pattern. The **v3_voice** ablation (geo base + Q18.1/Q19 only, without the transit dump) now has a greedy-decode answer: for Llama-3.1 it yields group MAE **5.64** and IP MAE **5.82** — better than the kitchen-sink **transit** (6.07 / 7.77) and **full** tiers with **no** interpersonal penalty, though IP remains above the geo base (4.63). Open-text voice is therefore not the driver of the transit collapse; the bundled structured mobility dump is.

ML versus live LLM on shared metrics (full-cohort vLLM)

The mock-LLM pipeline diagnostics rendered in earlier revisions are removed. Every table in this section is computed from the **committed full-cohort vLLM exports** (`exports/v1/`, `exports/v2/`, `exports/v3/`) and the **seeded tabular suite** (tables committed in `artifacts/posit_full_cohort/ml_vs_llm/`); no deterministic-mock predictions are reported. The best live agent is ******DeepSeek-R1-Distill-Llama-8B**** under v2 packaging** (pooled group MAE 5.02, IP 5.26); Random Forest (`random_forest`) is the classical reference. Full write-up: [memos/feature_predictive_power_ml_llm.qmd](#); head-to-head framing: [docs/ml_vs_llm.md](#).

MAE and band F1 by tier

Mean of group + interpersonal targets ($F1$ = macro-averaged harmonic mean of precision and recall; table: `metrics_ml_llm.csv`):

Table 14. RF versus DeepSeek v2 MAE and band metrics by persona tier.

Tier	RF MAE ↓	DeepSeek v2 MAE	RF macro-F1 ↑	DeepSeek v2 macro-F1	RF band acc	DeepSeek v2 band acc
demos	5.47	5.08	0.314	0.272	0.367	0.342
employ- ment	5.35	5.47	0.312	0.250	0.369	0.342
geo	4.75	5.05	0.386	0.281	0.452	0.349
transit	4.48	5.03	0.422	0.265	0.506	0.360

RF improves **monotonically** as tiers accumulate (MAE $5.47 \rightarrow 4.48$; macro-F1 $0.31 \rightarrow 0.42$; band accuracy $0.37 \rightarrow 0.51$). DeepSeek v2 does not: **employment** is its *worst* tier (5.47), **transit** (5.03) barely beats **demos** (5.08), and its band metrics are flat at ~ 0.25 – 0.36 — below RF at every tier (Figure 3). In variance terms, the best tabular learner explains only a small share of true CA: RF R^2 rises from -0.37 – -0.31 at **demos** (group / interpersonal) to **0.06**–**0.10** at **transit** (RMSE 5.82 / 5.51; table: `metrics_ml_llm.csv`). Even at its richest tier the classical suite leaves the large majority of individual CA variance unexplained, which contextualizes both the modest $d = -0.46$ transit-group CA contrast above and the single-digit exact-match ceiling of the persona agents.

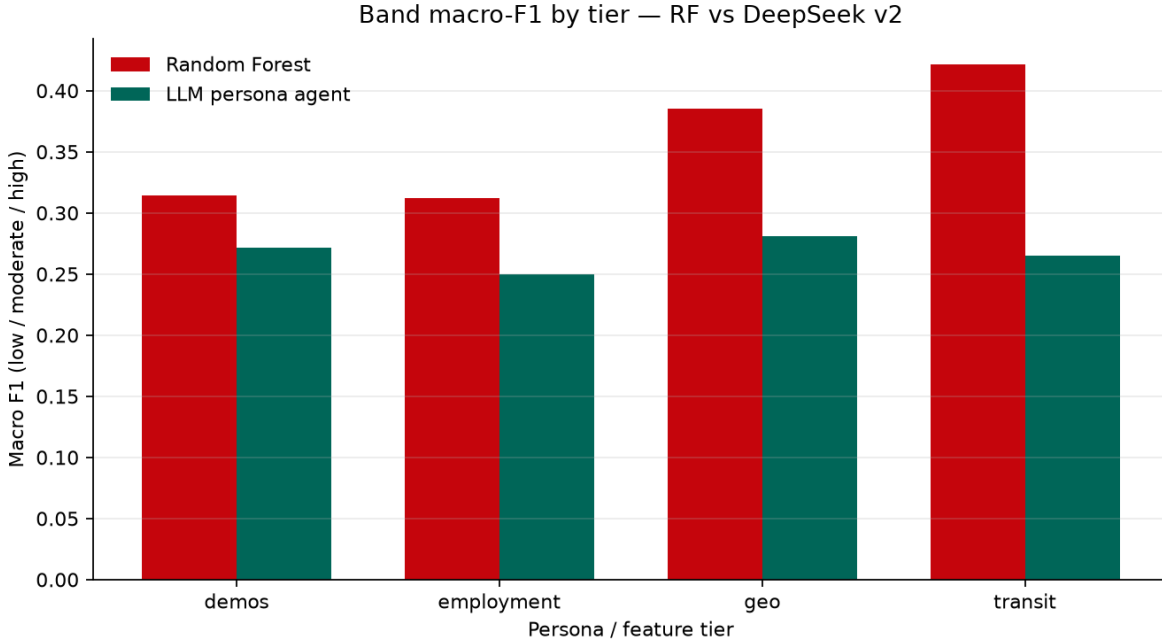


Figure 3. Band macro-F1 by tier — RF vs DeepSeek v2

Who uses the information? Tier ablations

Incremental Δ MAE (Group CA) when each feature group is added (table: `tier_ablation.csv`):

Table 15. Incremental change in Group-CA MAE as each persona tier is added.

Step	RF Δ MAE	DeepSeek v2 Δ MAE
+ employment	−0.12	+0.10
+ geo	−0.55	−0.26
+ transit	−0.37	+0.08

The tabular learner converts each added covariate group into lower error; the LLM’s profile is **non-monotonic** — employment and transit slightly *hurt* — matching its flat MAE across tiers and its higher **employment**-tier MAE above.

What the estimated bands actually contain

When a model says “low / moderate / high”, which numeric scores did it emit, and how far are those from ground truth? Group CA, **transit** tier (table: `band_score_profile_ml_llm.csv`):

Table 16. Predicted-score profiles inside each estimated band for Group CA at the transit tier.

Agent	Predicted band	n	Pred M (range)	GT M	MAE	Band precision
RF	low (6–13)	96	11.95 (8.56–13.48)	12.10	3.49	0.75
RF	moderate (14–19)	133	15.92 (13.50–19.47)	16.30	5.31	0.31
RF	high (20–30)	12	21.12 (19.82–24.64)	16.17	7.17	0.25
DeepSeek v2	low (6–13)	78	11.18 (6–13)	13.76	5.45	0.56
DeepSeek v2	moderate (14–19)	162	15.01 (14–18)	15.06	4.73	0.26
DeepSeek v2	high (20–30)	0	—	—	—	—

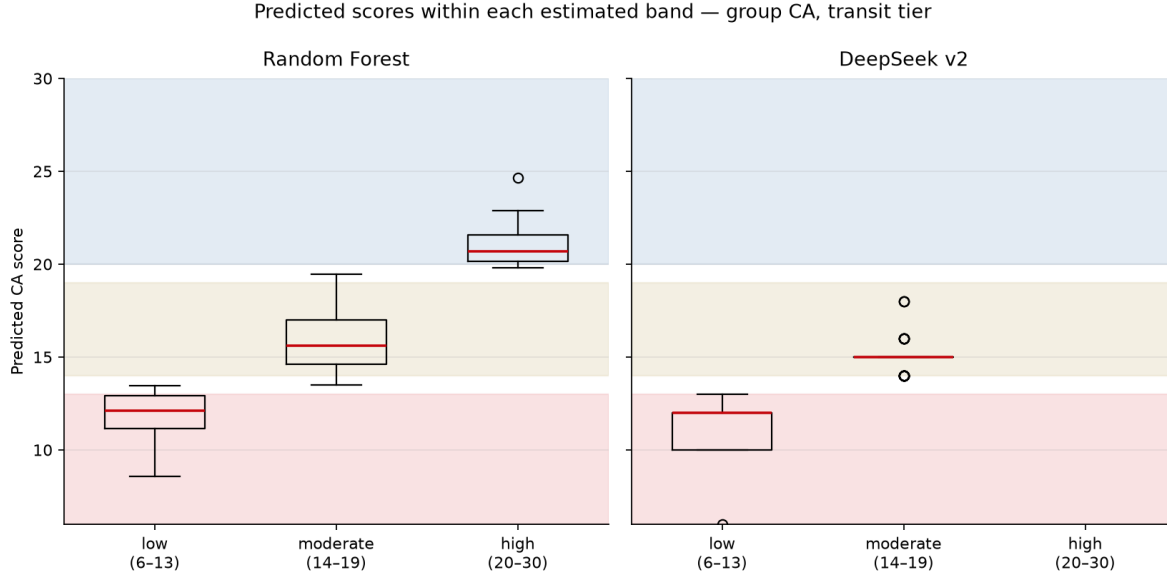


Figure 4. Predicted scores inside each estimated band — RF vs DeepSeek v2

! Important

Two quantification findings. **First, DeepSeek v2 never emits the high band for Group CA at transit** (0 of 240 predictions = 20) — its “moderate” label absorbs every high-apprehension respondent (Figure 4). **Second, RF’s rare high-band predictions ($n = 12$) are badly calibrated:** they land on respondents whose true mean is 16.17 (moderate), so RF precision on high is only 0.25. Both families are strong only on the low band.

Band discrimination: accuracy, distance, low-vs-high AUC

Disentangling MAE from discrimination: DeepSeek v2 is only $1.1\times$ worse than RF on MAE, but on band separation it is much worse — its continuous scores do **not** rank apprehension (table: `band_discrimination_ml_llm.csv`).

Table 17. Band discrimination at the `transit` tier: accuracy, distance, and AUC low-vs-high.

Agent	Tier	Side	Band acc	F1 macro	Mean band dist	AUC low-vs-high	Ordinal AUC
RF	transit	Group	0.481	0.386	0.573	0.717	0.650
RF	transit	Interpersonal	0.531	0.458	0.515	0.738	0.664

Agent	Tier	Side	Band acc	F1 macro	Mean band dist	AUC low-vs-high	Ordinal AUC
DeepSeek v2	transit	Group	0.358	0.267	0.704	0.527	0.518
DeepSeek v2	transit	Interpersonal	0.362	0.263	0.683	0.488	0.491

Notes. AUC low-vs-high = predicted score separating ground-truth high (20) from low (13) respondents (moderates excluded); ordinal AUC = Hand-Till pairwise AUC over all three bands (0.5 = chance). Full per-tier/per-band table: `artifacts/posit_full_cohort/ml_vs_llm/band_discrimination`

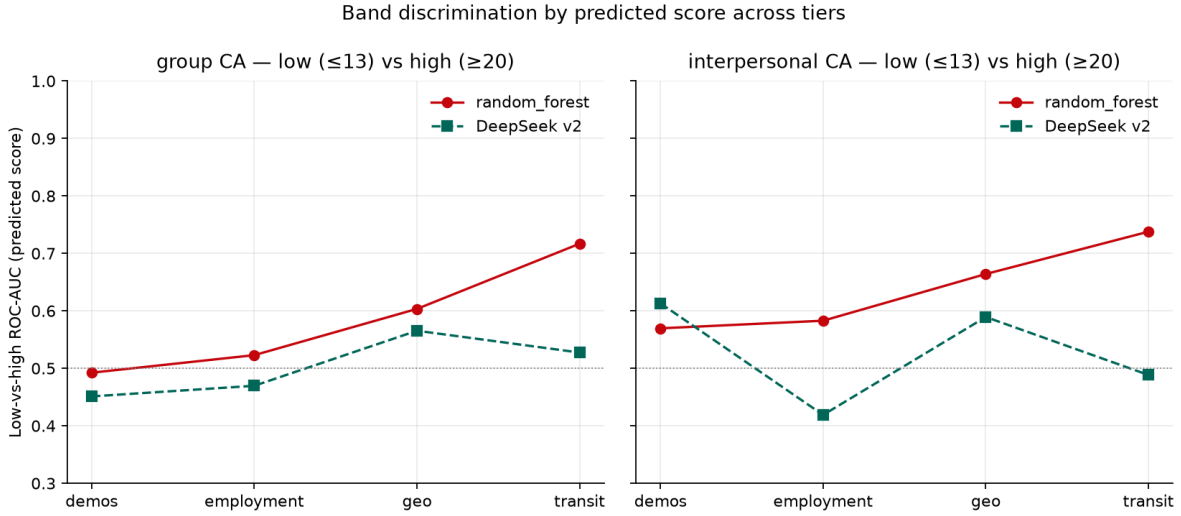


Figure 5. Low-vs-high ROC-AUC by tier — RF vs DeepSeek v2

! Important

RF’s low-vs-high AUC climbs from 0.49 (demos) to **0.72–0.74** (transit); DeepSeek v2 stays near chance (0.45–0.53) at every tier (Figure 5). The LLM’s error is **band-insensitive**: its scores hug the cohort mean, so absolute error is bounded while rank recovery fails. **MAE alone therefore understates the LLM’s failure to recover the trait distribution** — band F1, band distance, and low-vs-high AUC belong alongside MAE in digital-twin evaluation.

Feature attributions: TreeSHAP on true CA vs SHAP on live LLM outputs

TreeSHAP mean |SHAP| at the **transit** tier (predicting true Group CA) versus SHAP on the live LLM’s outputs — an RF fit to DeepSeek v2’s real vLLM predictions (surrogate $R^2 = 0.74$; tables: `shap_ml_group_raw.csv`, `shap_llm_surrogate_group_raw.csv`):

Table 18. Top-six TreeSHAP attributions for true Group CA versus SHAP on DeepSeek v2’s live outputs.

Rank	RF → true CA	mean SHAP	DeepSeek v2 output	
1	Q28 (ride-share days)	1.93	LocationLatitude	0.27
2	Employment status	0.83	Age	0.21
3	Q26 (public-transit days)	0.58	LocationLongitude	0.21
4	LocationLatitude	0.57	Q28	0.19
5	LocationLongitude	0.54	Country of residence	0.15
6	Age	0.53	Q26	0.15

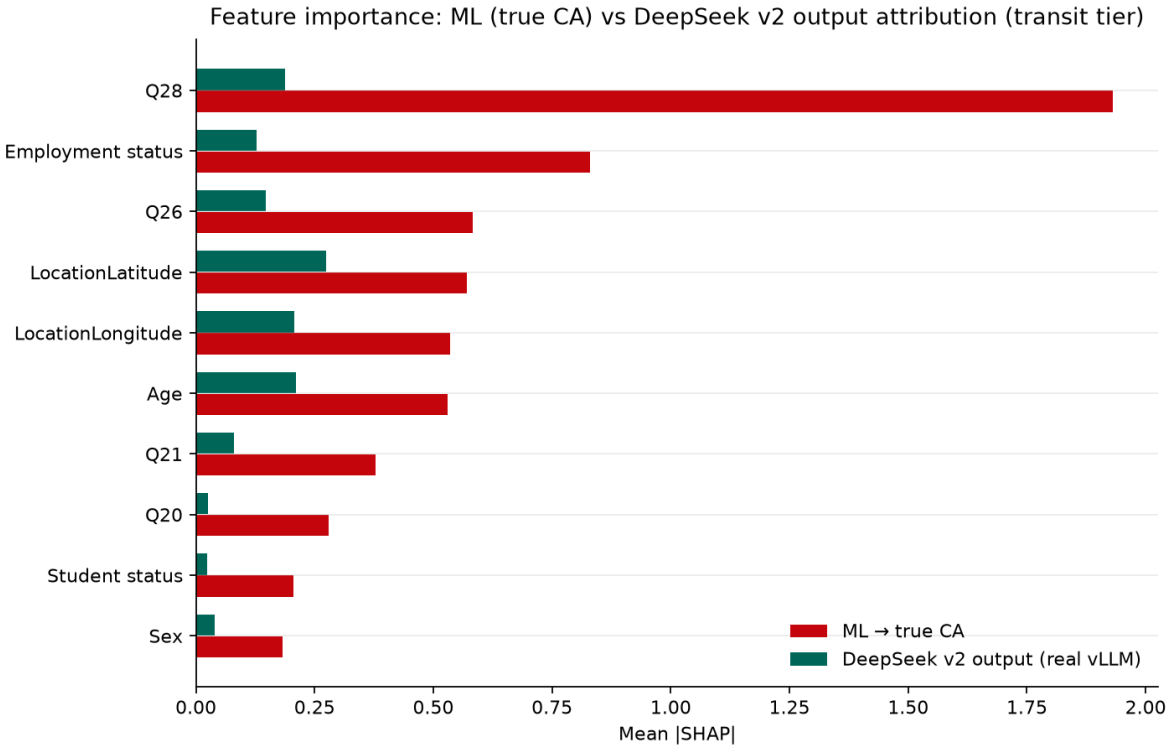


Figure 6. ML TreeSHAP vs DeepSeek v2 output SHAP

Ride-share days (Q28) and employment drive true-CA prediction, while the persona agent’s

live outputs are dominated by **geolocation and age** — over-weighting place/demographic cues relative to the mobility/employment mix that predicts the actual PRCA scores (Figure 6). Mean |SHAP| magnitudes are also approximately $7\times$ smaller for the LLM output attribution, so no single profile feature dominates the agent’s scores.

Error by demographic group (live DeepSeek v2)

Before interpreting group-level error gaps, it is useful to anchor them against the ground-truth CA distribution per group (committed `ground_truth_aggregates.csv`; group-CA M (SD)):

```
gt_agg = pd.read_csv(COMMITTED_FULL / "ground_truth_aggregates.csv")
rows = []
for _, r in gt_agg[gt_agg["scope"].isin(["sex", "student_status", "employment"])].iterrows():
    label = r["group_key"] if str(r["group_key"]) != "nan" else "Student = Missing"
    rows.append(
        {
            "Slice": f"{label}",
            "n": f"{r['n']:.0f}",
            "Group CA *M* (*SD*)": f"{r['mean_group_ca']:.2f} ({r['std_group_ca']:.2f})",
            "Group band high (%)": f"{r['group_band_high_pct'] * 100:.1f}",
        }
    )
pd.DataFrame(rows)
```

Table 19. Ground-Truth Group-CA by Demographic Slice ($N = 241$)

	Slice	n	Group CA *M* (*SD*)	Group band high (%)
0	Female	120	14.88 (5.76)	21.7
1	Male	121	14.37 (6.26)	19.8
2	No	190	14.26 (5.99)	20.5
3	Yes	34	16.35 (6.11)	20.6
4	Student = Missing	17	15.24 (5.51)	23.5
5	Full-Time	148	13.32 (5.14)	12.8
6	Other	62	16.95 (6.42)	37.1
7	Part-Time	31	16.16 (7.22)	25.8

Note. Group-CA means from the full matched analytic cohort. Full-time employment and (to a lesser degree) the student-status differences below sit at the floor of the “error by group” table that follows: they document the *true* group-level CA structure that the live agent’s calibration split either tracks or misses.

Real stereotyping slices from the committed v2 export (`exports/v2/.../tables/stereotyping_by_*.csv`), group-CA MAE by tier:

Table 20. Group-CA MAE by demographic slice across v2 tiers for **DeepSeek-R1-Distill-Llama-8B**.

Group	demos	employment	geo	transit	n (transit)
Student = Yes	5.24	4.71	4.68	5.00	34
Student = No	5.07	5.23	4.95	4.89	190
Female	4.93	5.09	4.73	4.73	120
Male	5.17	5.21	5.04	5.20	120
Full-Time	4.37	4.63	4.35	4.39	148
Part-Time	6.06	5.84	6.00	5.55	31
Other	6.16	6.05	5.61	6.07	61

Sex and student-status MAE gaps are small (≤ 0.5 points). **Employment is the striking slice**: Full-Time error sits at 4.35–4.63 in every tier, while Part-Time and Other hover at 5.5–6.2. Signed error (predicted – true) is near zero for Full-Time (-0.01) but **-2.0 to -3.6 for Part-Time / Other** at every tier — the live agent systematically **under-predicts CA for non-full-time respondents** ([docs/stereotyping_evaluation.md](#)). The ground-truth slice above (Group CA $M = 13.3$ for Full-Time vs $16.2 / 17.0$ for Part-Time / Other; high-band prevalence 13% vs $26\% / 37\%$) shows this is **not** symmetric noise: non-full-time respondents genuinely cluster higher on CA, so the agent’s negative signed error is a real *calibration miss* on a subgroup with elevated trait levels, not just a scatter artifact. This sharpens the RQ1 story: employment does not *shrink* overall MAE, but it is a real **calibration split** in the live model.

Secondary results: regular transit and mobility predictors

Regular transit is defined as Q26 {4–8 days a month, 8 or more days a month} (weekly-or-more), yielding **101 regular / 140 not regular** riders (prevalence = **0.419**) in the analytic cohort. The analyses below use stratified 5-fold Random Forests (`random_state=42`) unless noted. Full write-ups: [memos/transit_riders_ca.qmd](#), [memos/geo_predicts_transit.qmd](#), [memos/ca_scores_predict_transit.qmd](#), [memos/q27_q28_predict_transit.qmd](#), [memos/transit_covariates.qmd](#).

Regular riders differ in CA scores

Welch tests on the matched cohort show lower CA among regular riders ([memo](#); Figure 7):

Table 21. Welch contrasts for regular versus non-regular public-transit riders on PRCA subscales.

Sub-scale	Regular <i>M (SD)</i>	Not regular <i>M (SD)</i>	Δ	Cohen's <i>d</i>	Welch <i>t</i>	<i>df</i>	<i>p</i>	95% boot- strap CI for Δ
Group	13.04	15.76	−2.72	−0.46	−3.68	236.91	< .001	[−4.15,
CA	(5.08)	(6.42)						−1.28]
Inter-	13.31	15.04	−1.73	−0.30	−2.34	228.80	.020	[−3.12,
per-	(5.37)	(6.04)						−0.25]
sonal								
CA								

Note. Cell sizes: *n* = 101 regular / 140 not regular.

```

g = secondary["transit_ca"]["group"]
i = secondary["transit_ca"]["interpersonal"]
fig, ax = plt.subplots(figsize=(6.4, 4.0))
grouped_bars(
    ax,
    ["Group CA", "Interpersonal CA"],
    {
        f"Regular (n={g['n_regular']})": [g["mean_regular"], i["mean_regular"]],
        f"Not regular (n={g['n_not_regular']})": [g["mean_not_regular"], i["mean_not_regular"]],
    },
    ylabel="Mean PRCA score (6-30)",
    ylim=(0, 22),
)
fig.tight_layout()
plt.show()

```

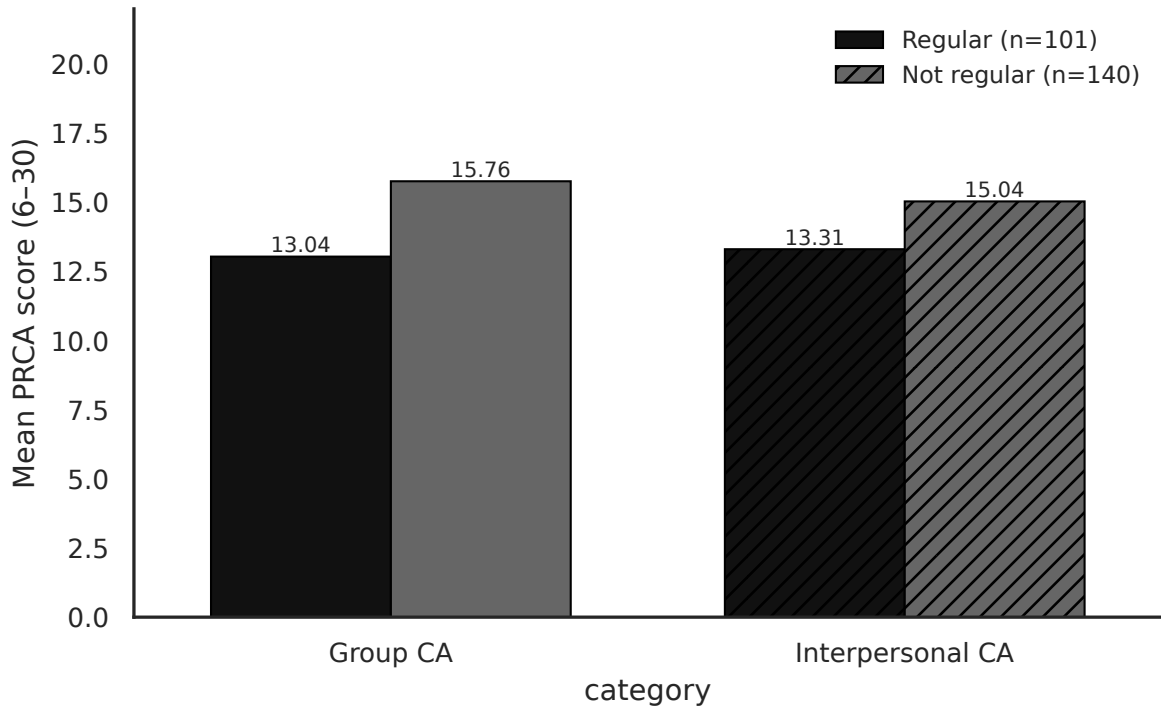


Figure 7. Mean Ground-Truth PRCA Scores for Regular Versus Non-Regular Public-Transit Riders

Note. Full matched analytic cohort. Means correspond to the Welch contrasts reported above and in [memos/transit_riders_ca.qmd](#).

Geography and CA as classifiers of regular transit

On the same outcome, a lat/long Random Forest yields cross-validated receiver-operating-characteristic AUC (CV ROC-AUC) = **0.551** (country-only AUC = **0.549**; chance = **0.500**; Figure 8; [memo](#)). A Random Forest on group + interpersonal CA yields CV ROC-AUC = **0.590** (group-only = **0.555**; interpersonal-only = **0.506**; out-of-fold confusion: true negatives = 76, false positives = 64, false negatives = 40, true positives = 61) ([memo](#)).

```
labels = [
  "CA: group + interpersonal",
  "CA: group only",
  "Geo: lat/long",
  "Geo: country only",
  "CA: interpersonal only",
```

```

    "Chance",
]
vals = [
    secondary["ca_rf"]["roc_auc"],
    secondary["ca_rf"]["group_only"],
    secondary["geo_rf"]["roc_auc"],
    secondary["geo_rf"]["country_only_roc_auc"],
    secondary["ca_rf"]["interpersonal_only"],
    0.5,
]
fig, ax = plt.subplots(figsize=(7.0, 3.8))
horizontal_bar(
    ax,
    labels,
    vals,
    xlabel="Cross-validated ROC-AUC",
    highlight_index=0,
    vlines={"Chance = 0.500": 0.5},
    xlim=(0.45, 0.68),
    legend_loc="below",
)
fig.tight_layout()
plt.show()

```

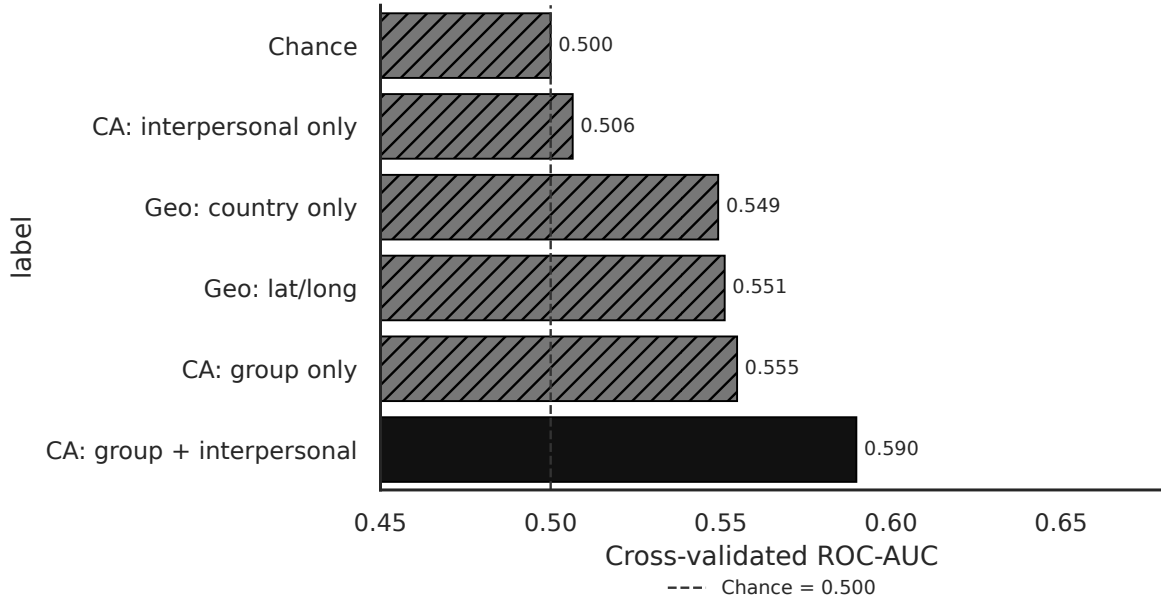


Figure 8. Cross-Validated ROC-AUC for Geography and Communication-Apprehension Random Forests Predicting Regular Transit

Note. Stratified 5-fold CV, `random_state=42`, $N = 241$. Detailed results: [memos/geo_predicts_transit.qmd](#); [memos/ca_scores_predict_transit.qmd](#).

Q27 and Q28 as traditional ML predictors of regular transit

Figure 9, Figure 10, and Table 22 summarize the Q27/Q28 reverse-prediction results ([memo](#)). **Q28** (ride-share days) shows a monotonic prevalence gradient from **15.7%** regular (Never) to **94.1%** regular (8+ days/month) and CV ROC-AUC = **0.762** (average precision [AP] = 0.689; balanced accuracy = 0.730; F1 = 0.702). **Q27** (rides on a typical transit day) yields CV ROC-AUC = **0.589**. The joint Q27+Q28 forest reaches AUC = **0.761** ($\Delta = -0.001$ vs Q28 alone).

```
order27 = [
  "1-2 rides in a typical day",
  "3-4 rides in a typical day",
  "5-6 rides in a typical day",
  "7 or more rides in a typical day",
]
order28 = [
  "Never",
```

```

    "0-1 days a month",
    "2-4 days a month",
    "4-8 days a month",
    "8 or more days a month",
]
short27 = {
    "1-2 rides in a typical day": "1-2 rides/day",
    "3-4 rides in a typical day": "3-4 rides/day",
    "5-6 rides in a typical day": "5-6 rides/day",
    "7 or more rides in a typical day": "7+ rides/day",
}
short28 = {
    "Never": "Never",
    "0-1 days a month": "0-1 days/mo",
    "2-4 days a month": "2-4 days/mo",
    "4-8 days a month": "4-8 days/mo",
    "8 or more days a month": "8+ days/mo",
}
q27 = pd.DataFrame(secondary["q27_prevalence"])
q27["level"] = pd.Categorical(q27["level"], categories=order27, ordered=True)
q27 = q27.sort_values("level")
q28 = pd.DataFrame(secondary["q28_prevalence"])
q28["level"] = pd.Categorical(q28["level"], categories=order28, ordered=True)
q28 = q28.sort_values("level")

fig, axes = plt.subplots(1, 2, figsize=(10.6, 4.8))
for ax, frame, title, short_map in [
    (axes[0], q27, "Q27 (n = 239)", short27),
    (axes[1], q28, "Q28 (n = 241)", short28),
]:
    y = np.arange(len(frame))
    ax.barh(y, frame["prevalence"], color="#444444", edgecolor="black", height=0.68)
    ax.axvline(
        secondary["prevalence"],
        color="black",
        ls="--",
        lw=1.1,
        label=f"Sample prevalence = {secondary['prevalence']:.3f}",
    )
    ax.set_yticks(y)
    ax.set_yticklabels(
        [f"{short_map.get(lv, lv)} (n={int(n)})" for lv, n in zip(frame["level"], frame["n"])]
    )

```

```

        fontsize=10,
    )
    ax.set_xlabel("Proportion regular transit (weekly+)", fontsize=10)
    ax.set_xlim(0, 1.14)
    for yi, v in zip(y, frame["prevalence"]):
        # Value labels always sit OUTSIDE the bar so they stay legible.
        ax.text(v + 0.012, yi, f"{v:.1%}", va="center", ha="left", fontsize=9.5)
    ax.invert_yaxis()
    apa_axes(ax)
    ax.tick_params(axis="x", labelsize=9)
    ax.text(0.985, 0.98, title, transform=ax.transAxes, ha="right", va="top", fontsize=10)
    ax.legend(frameon=False, fontsize=8, loc="upper right")
fig.tight_layout()
plt.show()

```

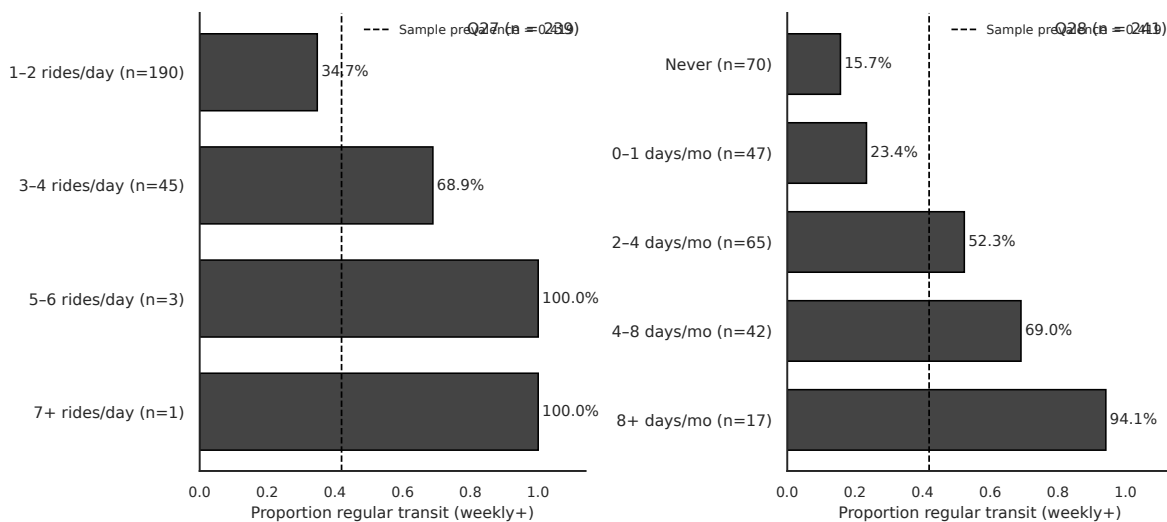


Figure 9. Proportion of Regular Transit Riders by Q27 Intensity and Q28 Ride-Share Days

Note. Dashed line marks cohort prevalence (0.419). Cell sizes for Q27 levels above 3–4 rides are sparse ($n = 3$ and $n = 1$). Source memo: [memos/q27_q28_predict_transit.qmd](#).

```

cmp = pd.DataFrame(secondary["covariate_comparison"])
keep = [
    "q28_days",
    "q27_q28",
    "rideshare",
    "ca_benchmark",

```

```

    "q27_intensity",
    "geo_benchmark",
    "chance",
]
labels = {
    "q28_days": "Q28 only",
    "q27_q28": "Q27 + Q28",
    "rideshare": "Q28 + Q29 (ride-share family)",
    "ca_benchmark": "Group + interpersonal CA",
    "q27_intensity": "Q27 only",
    "geo_benchmark": "Lat/long geo",
    "chance": "Chance / prevalence",
}
tab = cmp[cmp["spec_key"].isin(keep)].copy()
tab["spec_key"] = pd.Categorical(tab["spec_key"], categories=keep, ordered=True)
tab = tab.sort_values("spec_key")
out = pd.DataFrame(
    {
        "Model": tab["spec_key"].map(labels),
        "n": tab["n"],
        "ROC-AUC": tab["roc_auc"],
        "Average precision": tab["average_precision"],
        "Balanced accuracy": tab["balanced_accuracy"],
        "F1": tab["f1"],
    }
)
out = out.round(
    {
        "ROC-AUC": 3,
        "Average precision": 3,
        "Balanced accuracy": 3,
        "F1": 3,
    }
)
out

```

Table 22. Cross-Validated Random Forest Performance for Q27, Q28, and Benchmarks

	Model	n	ROC-AUC	Average precision	Balanced accuracy	F1
0	Q28 only	241.0	0.762	0.689	0.730	0.702
1	Q27 + Q28	239.0	0.761	0.690	0.696	0.667

Table 22. Cross-Validated Random Forest Performance for Q27, Q28, and Benchmarks

	Model	n	ROC-AUC	Average precision	Balanced accuracy	F1
3	Q28 + Q29 (ride-share family)	233.0	0.745	0.655	0.731	0.709
5	Group + interpersonal CA	241.0	0.590	NaN	NaN	NaN
6	Q27 only	239.0	0.589	0.513	0.623	0.467
7	Lat/long geo	241.0	0.551	NaN	NaN	NaN
9	Chance / prevalence	NaN	0.500	NaN	0.500	NaN

```

plot_rows = out.dropna(subset=["ROC-AUC"]).copy()
fig, ax = plt.subplots(figsize=(7.2, 4.2))
horizontal_bar(
    ax,
    plot_rows["Model"].tolist(),
    plot_rows["ROC-AUC"].tolist(),
    xlabel="Cross-validated ROC-AUC",
    highlight_index=0,
    vl_lines={"Chance = 0.500": 0.5, "Geo = 0.551": 0.551, "CA = 0.590": 0.590},
    xlim=(0.45, 0.84),
    legend_loc="below",
)
fig.tight_layout()
plt.show()

```

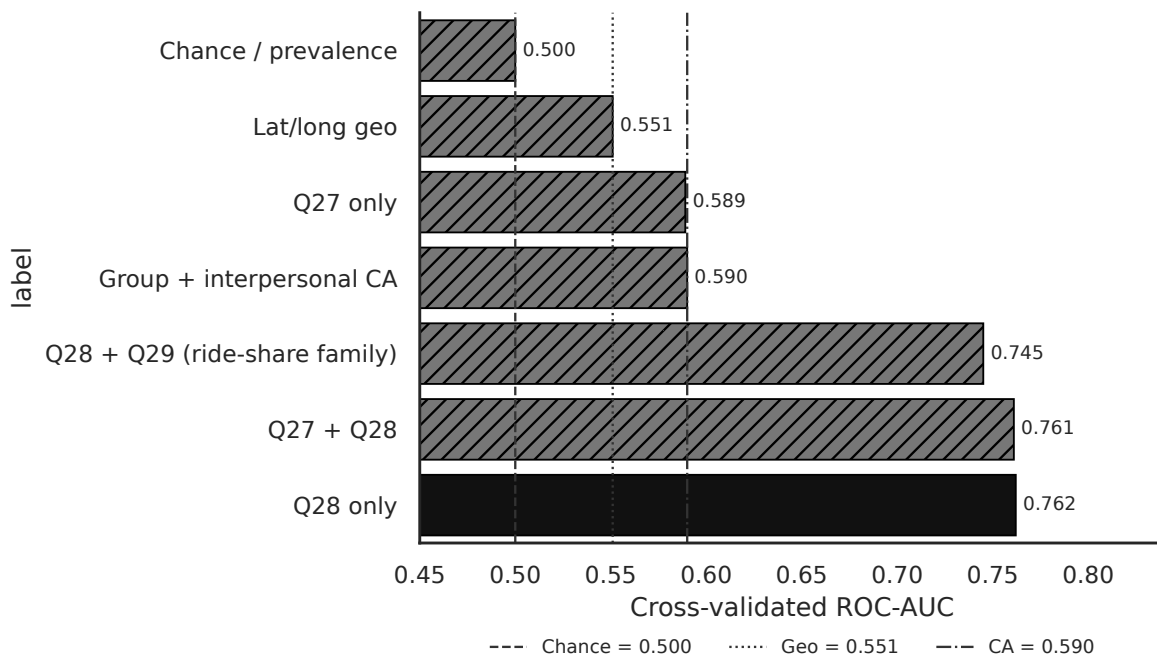



Figure 10. Cross-Validated ROC-AUC for Q27, Q28, Joint, and Benchmark Models Predicting Regular Transit

Note. Stratified 5-fold CV, `random_state=42`. Q28 alone exceeds chance by +0.262 AUC, geography by +0.211, and CA scores by +0.172. Head-to-head mobility follow-ups (car access AUC = 0.607; employment AUC = 0.528; rideshare family AUC = 0.745) are in [memos/transit_covariate_followups.qmd](#), [memos/car_access_predicts_transit.qmd](#), [memos/employment_predicts_transit.qmd](#), and [memos/rideshare_predicts_transit.qmd](#).

Extended secondary results: wave-2 follow-up experiments

Wave-1 memos left concrete open questions (unused Prolific demographics; country vs coordinates; Q28 after car conditioning; psych vs mobility importance; `country×car`; Q27 within regular riders; fair common- N rankings; residual CA after Q28). Eight offline experiments address those gaps with the same matched cohort and RF / Welch toolkit (`ca-personas followup-experiments --seed 42`). The agenda and methods hub are [docs/research_memo_agenda.md](#) and [docs/secondary_rq_followup_experiments.md](#).

Overview of seeded results

Wave-2 follow-up experiments — discrimination overview

Best: Multiply-imputed vs complete-case head-to-head (AUC=0.762, n=241) · Green ≥ 0.70 · Teal ≥ 0.60 · Slate ≥ 0.55 · Tan < 0.55 · Reference lines: chance / geo / CA / full-cohort Q28

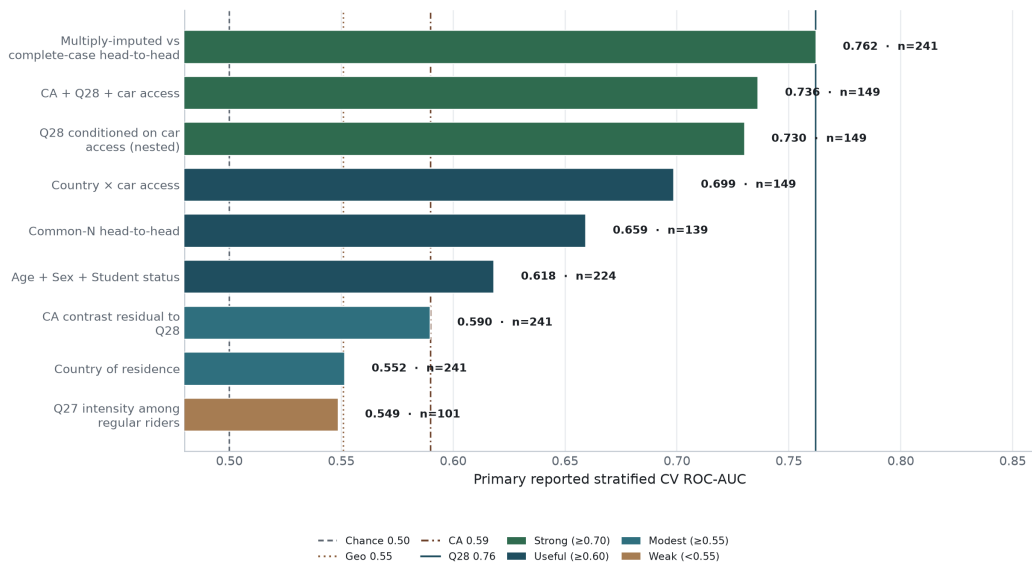


Figure 11. Cross-validated ROC-AUC across the eight wave-2 follow-up experiments

Table 23. Headline CV ROC-AUC for the eight wave-2 follow-up experiments.

Experiment	Memo	<i>n</i>	ROC-AUC	Takeaway
CA + Q28 + car (joint)	memo	149	0.736	Mobility-dominated joint model
Q28 car (nested)	memo	149	0.730	Q28 retains lift; car adds +0.066 AUC
Country + car	memo	149	0.699	Place × auto access synergy
Common- <i>N</i> best (Q28)	memo	139	0.659	Rideshare ranking robust to equal <i>N</i>
Demographics (Age+Sex+Student)	memo	224	0.618	Age-driven modest signal (younger 57.1% vs older 24.6% regular)

Experiment	Memo	n	ROC-AUC	Takeaway
Residual CA (CA-only arm)	memo	241	0.590	CA-only AUC; joint CA+Q28 = 0.783 (+0.021 over Q28)
Country only	memo	241	0.552	Ties lat/long geo
Q27 intensity among riders	memo	95–101	0.549	No useful within-rider classifier

Findings that change interpretation of the transit tier

1. **Q28 is not just “no car.”** On the car-complete overlap ($n = 149$), Q28 alone AUC = 0.665 and Q28+Q21 = 0.730—ride-share days retain discrimination after car conditioning, and car access adds +0.066 AUC ([nested memo](#)).
2. **Fair rankings still crown rideshare.** Forcing Q28, car, country, geo, employment, CA, and demographics onto one complete-case frame ($n = 139$) preserves **Q28 > country > car > geo > employment CA demos** ([common- \$N\$ memo](#)).
3. **Demographics matter for bias audits.** Age+Sex+Student reach AUC = 0.618 on $n = 224$; younger tertiles are far more often weekly+ riders (57.1% vs 24.6% older). That is relevant to whether **demos-tier** LLM errors track age stereotyping ([demographics memo](#)). Note this Age+Sex+Student feature set differs from the comprehensive-memo “demographics” ablation that also includes country (AUC = 0.636).
4. **CA is descriptively real, predictively redundant with Q28.** The overall regular-rider CA gap replicates (group $\Delta = -2.72$, $d = -0.46$), but nesting CA with Q28 lifts AUC only from 0.762 $\rightarrow$ 0.783. Within Q28 strata the CA gap is heterogeneous—including a sign reversal in the 0–1 days cell ([residual CA memo](#)).
5. **Q27 remains weak even as an among-rider outcome.** High intensity among weekly+ riders is not recoverable from CA, Q28, car, employment, or demos (best AUC = 0.549).
6. **Kitchen-sink ceiling still looks like Q28.** A tuned comprehensive forest over demos + employment + geo + car + ride-share + CA reaches CV ROC-AUC = **0.824**, but permutation importance remains Q28-dominated ([comprehensive memo](#)).

Unified results: one cohort, three question families

The primary (RQ1–RQ3), secondary (S1–S5), and focus (TF1–TF2) analyses share the same matched analytic cohort ($N = 241$) and the same information-tier logic, so they can be read as one measurement story. Table 24 collects each question’s headline statistic and bottom line.

Table 24. Unified results across the primary, secondary, and focus question families.

Question	Headline result	Bottom line
RQ1 · employment	v1 tier-level lift negligible; live DeepSeek v2 slice: Other MAE 6.07 vs Full-Time 4.39 at transit	Employment is a calibration split , not an error shrinker
RQ2 · transit	Model-dependent; Llama-3.1 IP collapse (+3.51 MAE; signed $-2.21 \rightarrow +6.52$); DeepSeek flat	Bundled mobility cues are a hazard for some families
RQ3 · full voice	Best DeepSeek full IP 5.42; Llama-3.2 full group 5.29; v3 voice stable	Open-text voice helps some models, hurts others
S1 · transit CA	Regular riders lower CA (group $\Delta -2.72$, $d -0.46$, $p < .001$)	CA and transit are descriptively related
S2 · geo $\rightarrow$ transit	lat/long RF CV AUC 0.551 (country 0.549)	Geography alone chance
S3 · CA $\rightarrow$ transit	CA RF CV AUC 0.590 (group 0.555, IP 0.506)	CA is a weak reverse predictor
S4 · Q27/Q28 $\rightarrow$ transit	Q28 AUC 0.762 ; Q27 0.589; joint 0.761	Ride-share exposure is the dominant mobility cue
S5 · covariate follow-ups	Q28 retains lift after car (0.730); CA+Q28+car 0.736; kitchen sink 0.824	No covariate displaces Q28
TF1 · profile $\rightarrow$ transit	Demos+employment+geo AUC 0.662 (0.672 with CA)	Ridership recoverable from non-mobility profile
TF2 · transit intensity	Q27 intensity AUC 0.549; MAE 1.22 vs majority 0.805	Fine-grained intensity is not recoverable

Three cross-cutting conclusions follow.

1. **The tabular signal is concentrated in mobility, and ride-share is its best proxy.**
Q28 leads both predictive directions: it is the top TreeSHAP feature for true CA (mean |SHAP| 1.93, more than twice the runner-up) and the top reverse predictor of weekly+

transit (AUC 0.762). Geography and CA scores stay near chance as reverse predictors (0.551 / 0.590); Q27 intensity and the TF2 intensity target remain weak. The persona tiers with the most justified information content are therefore **transit** (Q28/Q26) and **geo** — the same tiers that move RF error most (-0.55 , -0.37 Δ MAE).

2. **The live LLM does not use that signal, and MAE alone hides how badly.**
The best non-collapsed agent (DeepSeek v2) trails RF on every tier and metric, shows a non-monotonic tier profile (employment and transit slightly hurt), near-chance low-vs-high band discrimination (0.49–0.53 vs RF 0.72–0.74), no high-band emissions for group CA, and surrogate SHAP attributions dominated by geolocation/age rather than the Q28/employment mix that predicts true CA. Its absolute error is within $1.1\times$ of ML — but only because its scores hug the cohort mean.
3. **Tier effects are not additive gifts; they are model- and target-dependent.**
Employment is a calibration split in the live model (Part-Time/Other under-predicted by 2.0–3.6 points), transit breaks Llama-3.1’s interpersonal recovery but not DeepSeek’s, and voice text helps Llama-3.1’s group bands while hurting its IP error. Reporting a single pooled MAE per model therefore obscures the patterned failures that matter for digital-twin evaluation; band F1, band distance, low-vs-high AUC, and group-level error slices are the load-bearing metrics.

Discussion

Answers to the research questions

RQ1 — employment. Adding employment status does **not** shrink overall prediction error relative to **demos** (v1 tier-level Δ MAE = 0 across models), so we retain the null for the aggregate comparison. The alternative is nonetheless supported in a different form: live DeepSeek v2 shows a real **calibration split** — Full-Time respondents are under-approximated (MAE 4.35–4.63; signed error -0.01) while Part-Time/Other are systematically under-predicted (signed error -2.0 to -3.6) at every tier. Employment is therefore a *calibration split*, not an error shrinker. ([Memo: demographics \$\rightarrow\$ transit](#) · [Memo: live-LLM stereotyping slices](#))

RQ2 — transit. We reject the null for the LLM family in a *model-dependent* direction: the mobility bundle slightly helps Llama-3.1’s group bands but produces a **transit-triggered interpersonal collapse** (IP MAE $4.67 \rightarrow 8.17$; signed error flips $-2.21 \rightarrow +6.52$), while DeepSeek’s IP error stays flat. v3 greedy ablations isolate the trigger to the *bundled* mobility dump (bundled IP 7.77 vs single-cue 4.85–5.92). The classical RF baseline does improve monotonically with transit cues (MAE $5.47 \rightarrow 4.48$), so the null is rejected for the tabular family but the LLM uses mobility cues as stereotype proxies rather than psychologically sensible signal. ([Memo: Llama-3.1 v1 baseline](#) · [Memo: v2/v3 evaluation](#) · [Memo: cross-model comparison](#))

RQ3 — full voice. We reject the null only for specific model–target pairs: DeepSeek’s best full IP MAE (5.42), Llama-3.2’s best full group MAE (5.29). For Llama-3.1, voice partially walks back the transit collapse (IP 8.17 $\rightarrow$ 6.92) but stays above demos. Open-text voice is **not uniformly helpful** — no consistent aggregate improvement. ([Memo: cross-model comparison](#) · [Memo: v2/v3 evaluation](#))

S1 — transit CA. We reject the null: regular riders report lower group ($\Delta = -2.72$, $d = -0.46$, $p < .001$) and interpersonal ($\Delta = -1.73$, $d = -0.30$, $p = .020$) CA. ([Memo: transit riders vs CA](#))

S2 — geo $\rightarrow$ transit. We retain the null: lat/long RF CV ROC-AUC = 0.551 (country 0.549), statistically chance. ([Memo: geography \$\rightarrow\$ transit](#))

S3 — CA $\rightarrow$ transit. We retain the null in practical terms: CA RF ROC-AUC = 0.590, a weak reverse predictor well below conventional discriminability. ([Memo: CA \$\rightarrow\$ transit](#))

S4 — Q27/Q28 $\rightarrow$ transit. We reject the null for ride-share: Q28 ROC-AUC = **0.762** (AP 0.689; F1 0.702). Q27 intensity alone is chance (0.589), and adding Q27 to Q28 adds nothing (0.761). ([Memo: Q27/Q28 \$\rightarrow\$ transit](#))

S5 — follow-ups. We reject the null in a qualified form: wave-2 covariates add lift only when they extend the ride-share family (CA+Q28+car 0.736; kitchen sink 0.824), but no covariate displaces Q28’s dominance. ([Memo: covariate follow-ups](#) · [Memo: MI head-to-head](#))

TF1 — profile $\rightarrow$ transit. We reject the null: demos+employment+geo reaches AUC **0.662** (0.672 with CA) — ridership is partly recoverable from non-mobility profiles.

TF2 — transit intensity. We retain the null: fine-grained intensity is not recoverable (Q27 intensity AUC 0.549; Q26 ordinal MAE 1.22 vs majority 0.805). No answer is an answer here: transit frequency beyond weekly+ is effectively irreducible from the available covariates. ([TF1/TF2 memo](#))

Primary digital-twin findings

On the full matched analytic cohort ($N = 241$), each participant yields five persona prompts (demos $\rightarrow$ employment $\rightarrow$ geo $\rightarrow$ transit $\rightarrow$ full) with fixed ground-truth PRCA scores. Full-cohort **prompt-v1** vLLM exports for four open-weight models show that none is a high-fidelity digital twin (exact match 6–9%), and all remain above the classical ML suite floor of **4.49** group MAE / **4.25** interpersonal MAE at transit ([docs/ml_baselines.md](#)). Among non-collapsed runs, **DeepSeek-R1-Distill-Llama-8B** minimizes pooled group MAE (**5.22**) with the most tier-stable profile; **Llama-3.2-3B-Instruct** leads interpersonal MAE (**5.35**) and group band accuracy (**52.7%**); **Llama-3.1-8B-Instruct** recovers interpersonal bands early but **collapses at transit** (IP MAE 4.67 $\rightarrow$ **8.17**); **Llama-3.3-70B** is a mode-collapse cautionary case, not a scale win ([memos/vllm_v1_cross_model_comparison.qmd](#); Figure 1). Employment (RQ1) is negligible as a *tier-level* MAE mover across models, though the live

DeepSeek v2 slices reveal a real **calibration split** (Part-Time/Other under-predicted by 2.0–3.6 points; see Results); transit text (RQ2) is a **model-dependent hazard** rather than a universal aid. The archived v2/v3 exports refine this picture: DeepSeek improves under signal-first v2 packaging (group MAE **5.02**, IP **5.26**, transit IP 5.09), while v2 does **not** fix Llama-3.1’s collapse and even raises its base-tier IP error, and greedy v3 ablations show the collapse is **combination-specific** (single-cue tiers 4.85–5.92; bundled dump 7.77).

Why transit breaks Llama-3.1 but not DeepSeek

The Llama-3.1 interpersonal collapse is a signed-error flip ($-2.21 \rightarrow +6.52$), not random scatter (Figure 2). A plausible account — without claiming a causal architecture proof — is that **instruct-tuned** Llama-3.1 over-weights mobility language as a social-anxiety stereotype once the prompt-v1 transit dump appears, whereas the **reasoning-distilled** DeepSeek sibling keeps IP MAE flat and signed error near zero at the same tier (Cheng et al., 2023; Santurkar et al., 2023). This is consistent with the broader impersonation literature, in which the same persona-conditioned generation that recovers developmental or expertise effects simultaneously exposes systematic group biases (Salewski et al., 2023). Training-data and decoding differences may also matter (DeepSeek’s post-`</think>` ingest still varied predictions; Llama-3.3-70B collapsed to a constant prior). The archived GPU runs add two causal clues. First, signal-first v2 packaging with a mobility anti-bleed clause does **not** remove the failure (transit IP 8.23 v1 8.17) and shifts Llama-3.1’s base interpersonal calibration (demos IP 4.67 $\rightarrow$ 8.45) — packaging is model-specific, not a universal cure, echoing findings that surface-level prompt formatting alone shifts model answers (Sclar et al., 2024). Second, greedy v3 single-cue ablations leave IP stable (v3_public_transit 4.85, v3_voice 5.82, v3_rideshare 5.92) while the bundled dump still collapses it (**7.77**) — the trigger is the **combination** of mobility cues in one transit bundle, not any single item ([docs/persona_prompt_versions.qmd](#); [docs/llm_v2_v3_enhanced_variants.md](#)).

Q28 dominance and persona design

Secondary analyses show that **Q28 ride-share days** dominate reverse prediction of weekly+transit (AUC **0.762**) while geo and CA stay near chance and Q27 intensity is weak ([memos/q27_q28_predict_transit.qmd](#)). Q28 is a **mobility exposure cue**, not a demographic label. For persona design that matters: stereotyping audits that slice only on Age/Sex/Student can miss bias that tracks ride-share or transit exposure (Cheng et al., 2023). The evaluation battery therefore includes **regular_transit** and Q28 alongside Sex / Student / Employment / Age tertiles, reporting max–min MAE gaps and Δ -gap vs **demos** ([docs/stereotyping_evaluation.md](#)). Wave-2 follow-ups strengthen the Q28 claim — it retains lift after car access and leaves CA with only marginal incremental AUC

([docs/secondary_rq_followup_experiments.md](#)). Treating ride-share as the high-value mobility tip (and isolating it in `v3_rideshare`) aligns persona prompts with the tabular importance ranking ([docs/factor_feature_importance.md](#)).

Implications for survey research

Even the best live model (DeepSeek under v2 packaging: group MAE **5.02**) remains roughly $1.1\times$ worse than Ridge regression on the same targets (~ 4.49), and exact twin recovery stays single-digit. That is a caution for **LLM-as-respondent** / **silicon-sampling** pipelines (Argyle et al., 2023; Park et al., 2024): fluent persona inhabitation does not automatically recover individual psychometrics, parse success is not twin success (70B mode collapse), and classical tabular ceilings belong in the evaluation kit (Argyle et al., 2023; Hu & Collier, 2024). Prompt packaging and model choice can change error *patterns* (transit hazard vs stability) without closing the ML gap. In the terms of the emerging LLM-simulation program — whether framed as *homo silicus* economic agents (Horton et al., 2023), cognitive-model finetuning (Binz & Schulz, 2024), or the broader transformation of social-science research pipelines (Grossmann et al., 2023) — our benchmark adds a concrete psychometric caution: a model that is useful for *coarse opinion simulation* may still be a poor *individual-level digital twin* on a validated trait inventory, and the two claims must be separated. All displayed statistics on this page come from committed full-cohort vLLM exports, seeded tabular runs, and committed artifact tables — there are no mock-LLM claims; the deterministic mock provider is reserved for offline pipeline smoke tests.

Limitations

Limitations include (a) same-wave observational designs for reverse-prediction AUCs, which cannot establish that ride-share causes weekly+ transit (or vice versa); (b) Prolific convenience-sample generalizability beyond the matched US-leaning online panel; (c) model and version drift — published numbers are pinned to specific Hugging Face checkpoints and the `20260726_*` / `20260728_*` / `20260729_*` / `20260730_*` export stamps; (d) keyless Cloud renders load committed full-cohort artifacts (vLLM export tables, seeded RF results, ML-vs-LLM metric tables) rather than re-running live inference, so a re-render does not refresh the exports; and (e) v2/v3 GPU coverage is now broad but not complete — v2 is evaluated on Llama-3.1/3.2/DeepSeek and v3 (greedy) on Llama-3.1/3.2-3B-Instruct/3.3-70B, yet the canonical `v3_enhanced` decode refresh, a v2/v3 run on the base 3B, and Llama-3.3-70B under the `large_model` preset remain pending, and the 8-tier v3 pooling mixes collapsed with stable tiers. Complete-case shrinkage for car-conditioned follow-ups further limits some wave-2 contrasts.

Future work

Priority GPU work is a **canonical v3 refresh under the v3_enhanced decode preset** (temp 0.3, seed 42, guided JSON, mobility anti-bleed system text) on Llama-3.1, DeepSeek, Llama-3.2, and Llama-3.3-70B, since the committed v3 packages are greedy-decode ablations (identical to the archived `prior_v3_greedy` runs). v2 coverage is complete except for the base 3B and 70B; DeepSeek v2 and the 8-tier greedy ablations are evaluated and reported above (`exports/v2/`, `exports/v3/`). Use `ca-personas stereotype-eval` on each live export so Sex/Age/Student/Employment **and** mobility-exposure MAE gaps are reported with tier-widening Δ s ([docs/llm_v2_v3_enhanced_variants.md](#)). Additional follow-ups: constrained decoding for the 70B collapse (`large_model` preset), and participant-level prediction histograms once raw exports are staged.

Conclusion

On a full matched Prolific Qualtrics cohort ($N = 241$), no open-weight persona agent came close to being a high-fidelity PRCA digital twin: every live model exceeded the classical suite floor, exact recovery stayed single-digit, and the best agent (DeepSeek under v2 packaging, group MAE 5.02) trailed Ridge by roughly $1.1\times$ while showing near-chance band discrimination and a systematic employment calibration split. The tabular story is starker and more interpretable — ride-share exposure (Q28) dominated prediction of weekly+ transit (AUC 0.762) and was the top feature for true CA, while geography, CA scores, and transit-day intensity added little. Together these findings argue that **persona prompting should be audited with band-level and attribution metrics, not MAE alone**, and that classical tabular learners remain the load-bearing benchmark for LLM-as-respondent claims. The memo scaffolding under [memos/](#) and [docs/](#) documents each finding with the open questions that motivated the next wave of analyses.

Reproducibility

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
pip install -e .
# Stage File A/B/C under ../sibling_data or CA_SIBLING_DATA=/tmp/sibling_data
ca-personas run --provider mock --join inner
ca-personas covariate-transit-rf --join inner --seed 42
python scripts/sync_posit_full_cohort_artifacts.py
python scripts/regenerate_ml_vs_llm_shap.py    # rebuild ML-vs-LLM tables/figures + artifact s
quarto render
```

For **Posit Connect Cloud** (Posit PBC, 2024b, 2024a): publish this repository (Burleson, 2026) with `_quarto.yml` as the primary file and `requirements.txt` present so Python dependencies install at build time. When private exports are absent, the manuscript loads committed full-cohort artifacts — participants, seeded secondary metrics, ML-vs-LLM metric tables (`artifacts/posit_full_cohort/ml_vs_llm/`), and the vLLM export tables including real stereotyping slices. From the project root, run `./publish.sh` (full analyses → render → Posit Connect publish) or `./publish.sh --skip-analysis --skip-render` to re-push the current `_site/`.

Source and author profile: github.com/Exios66/psych755-jjb · github.com/Exios66 (GitHub, 2024).

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